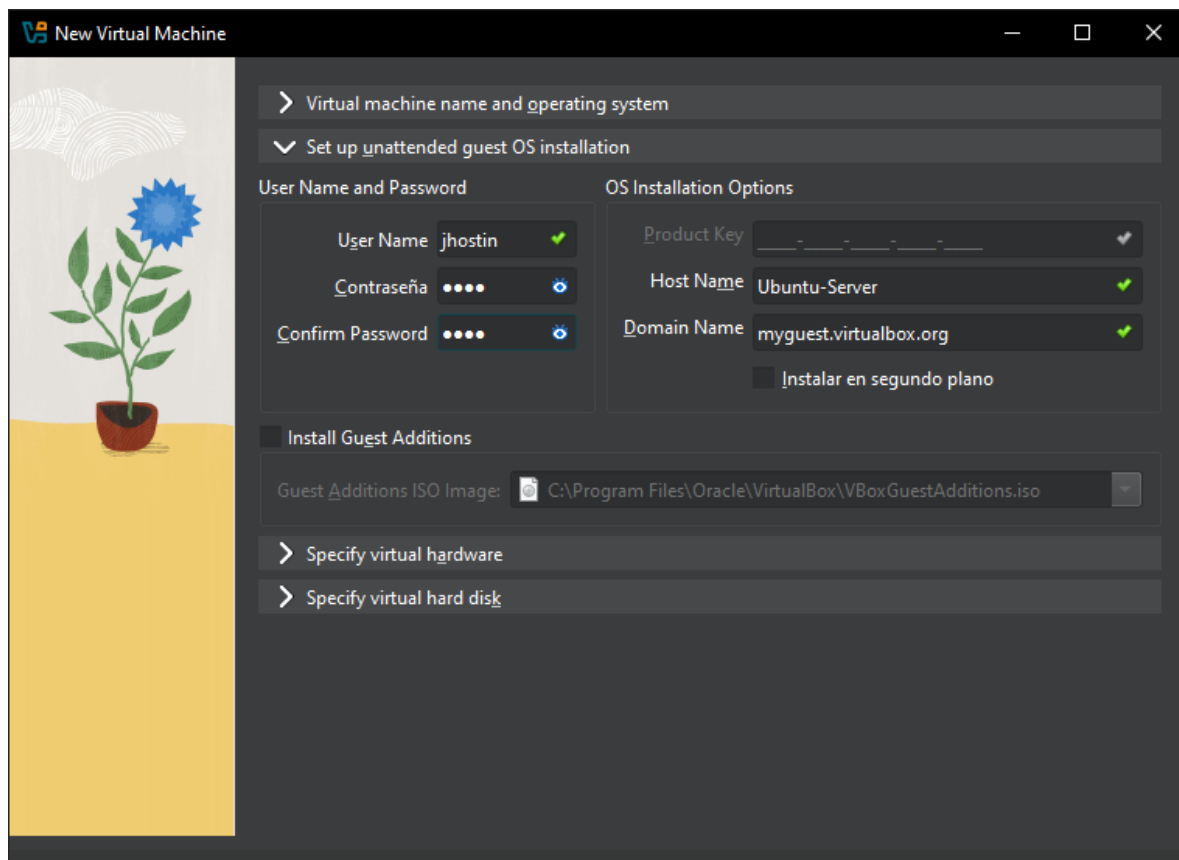
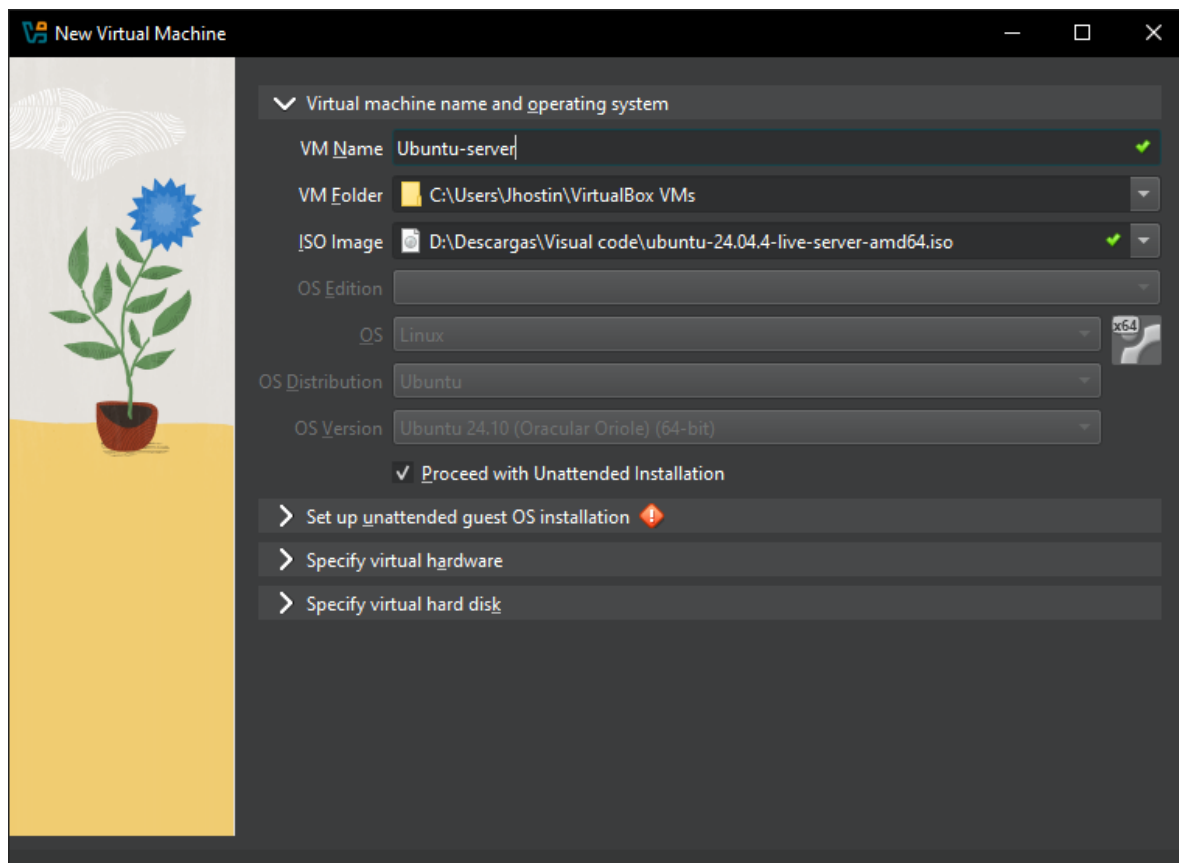
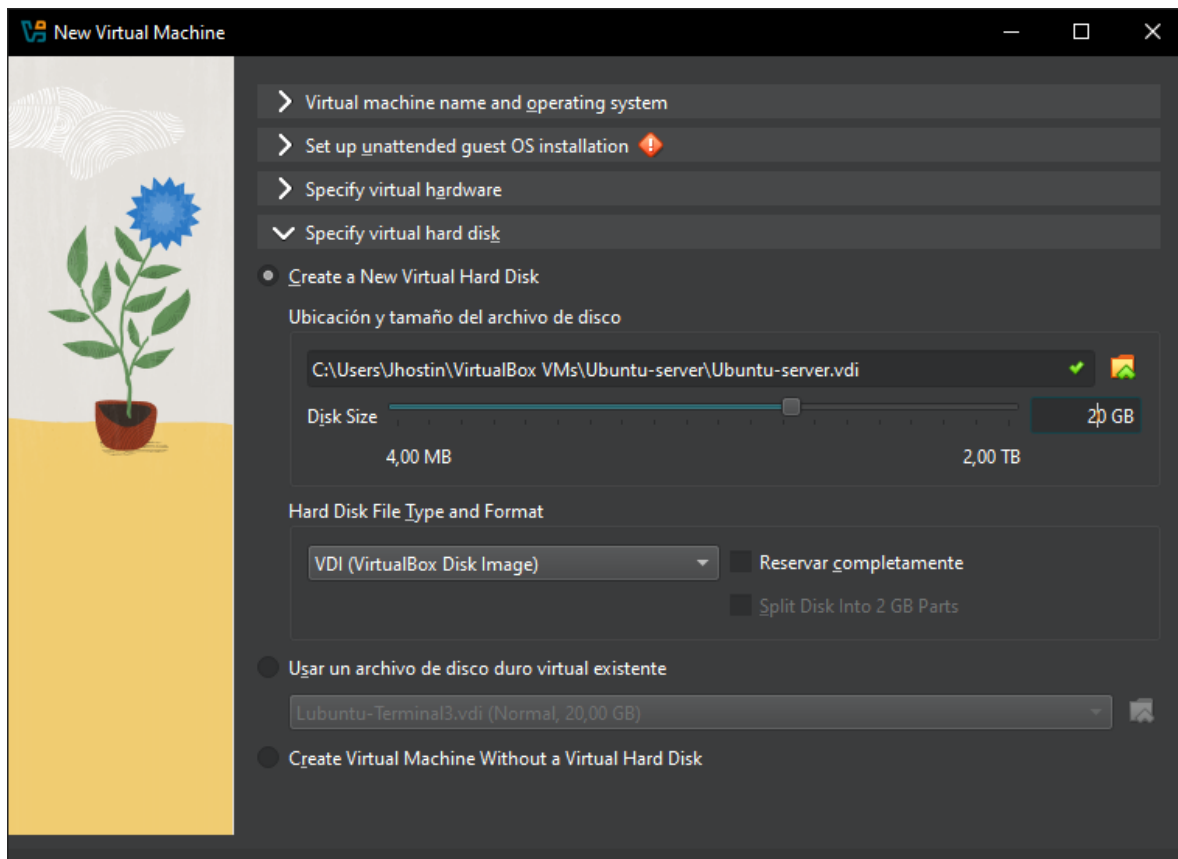
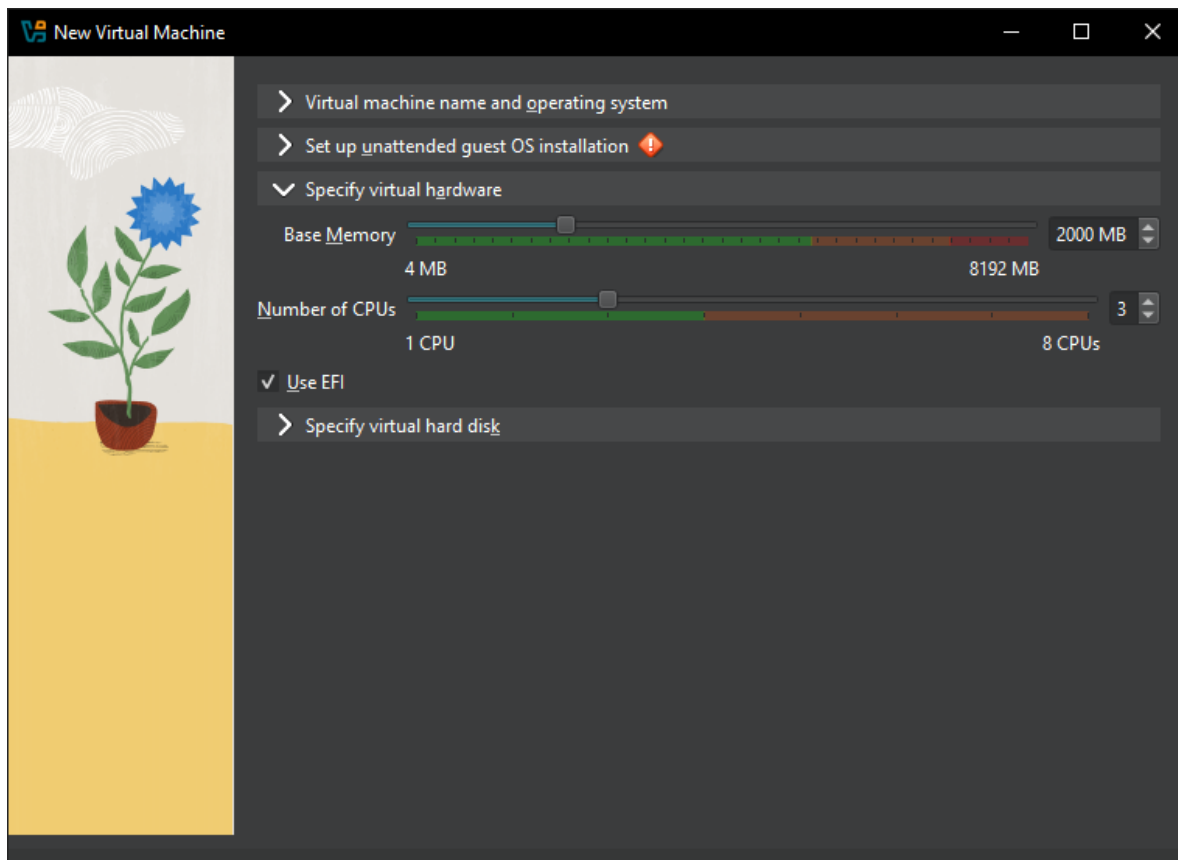


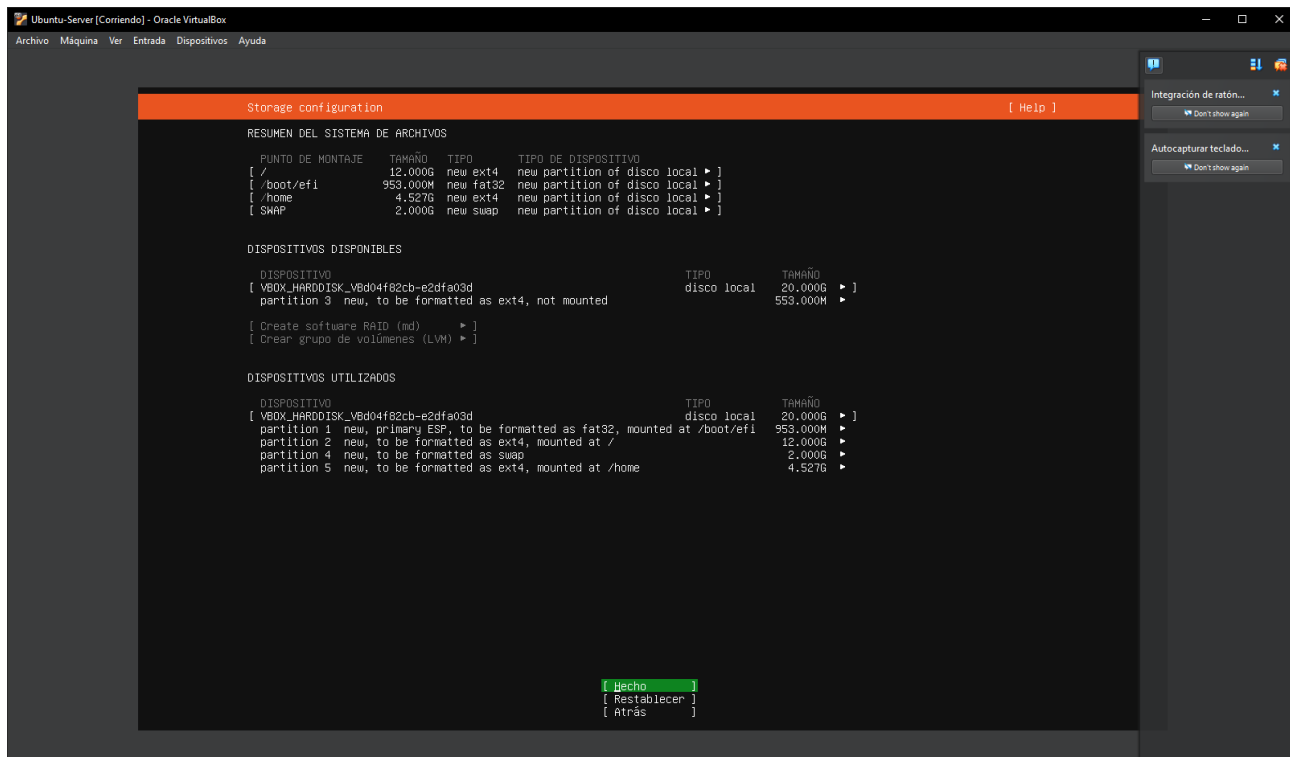
1. Virtualización con Linux

1.1 crear VM consola

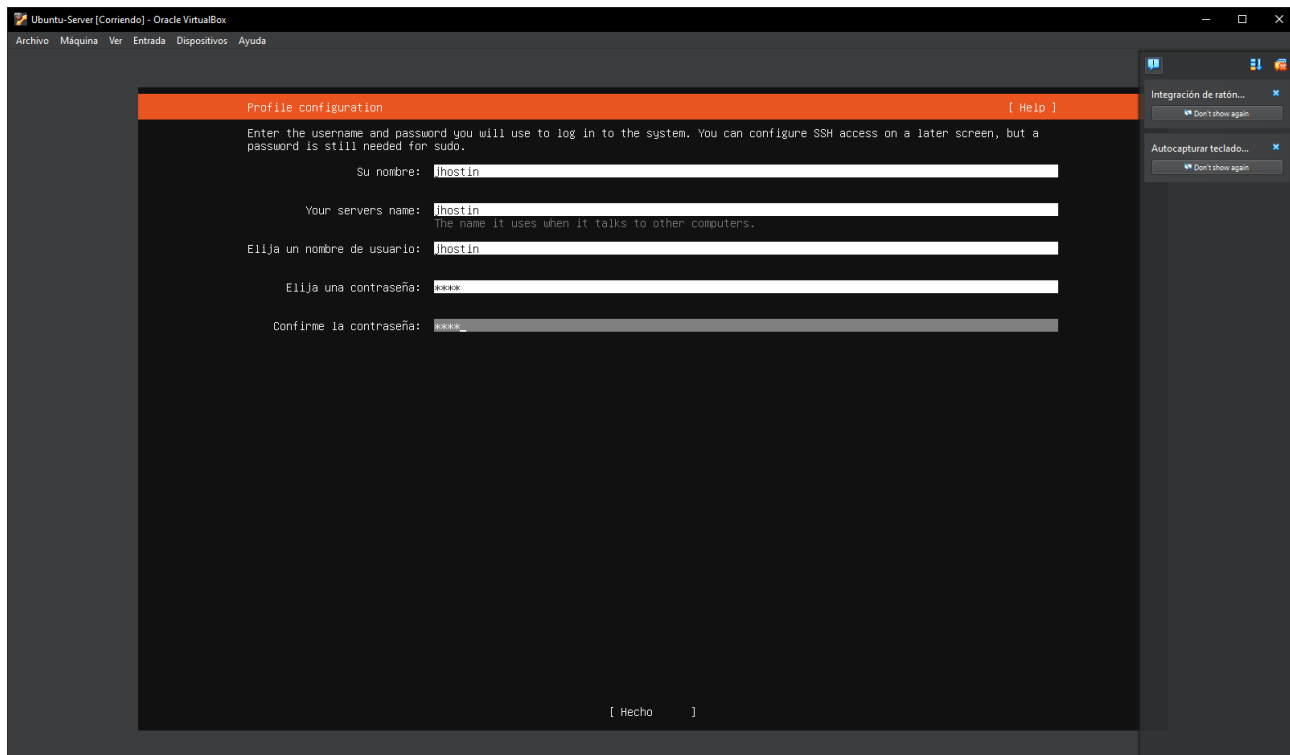




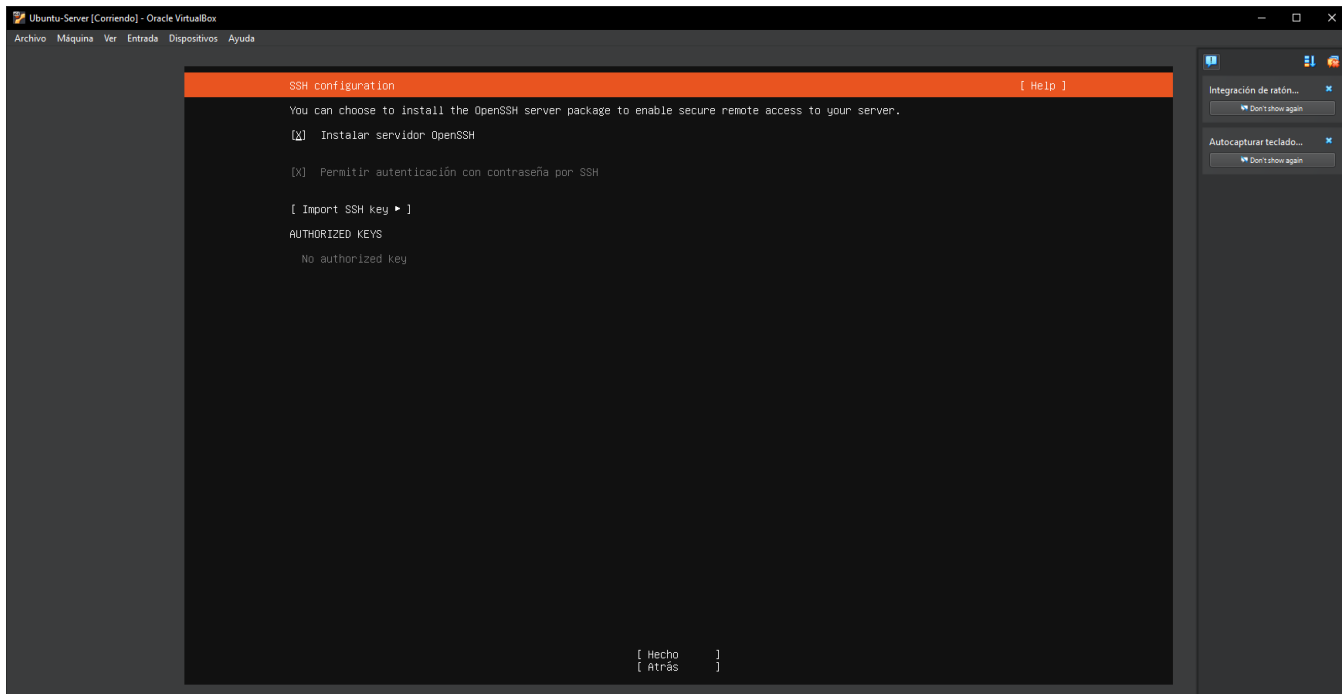
Creación de las particiones



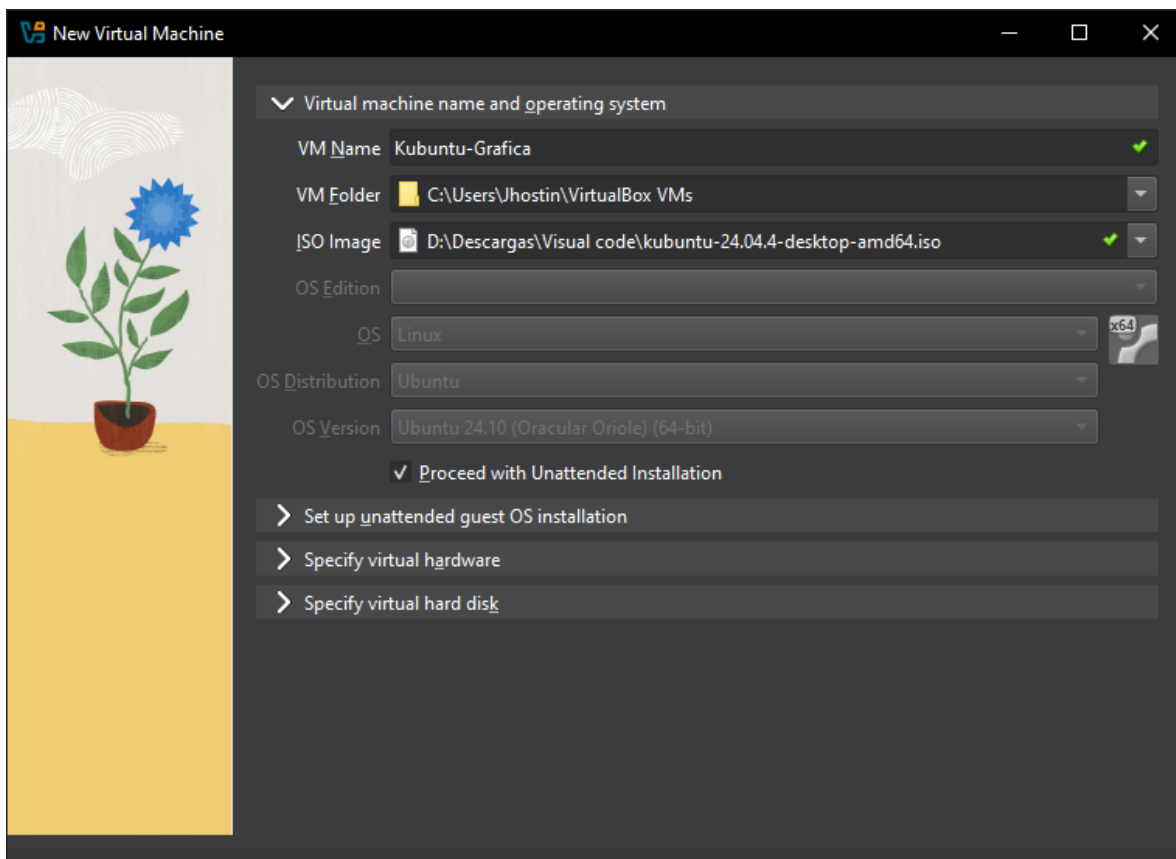
Creación de usuario

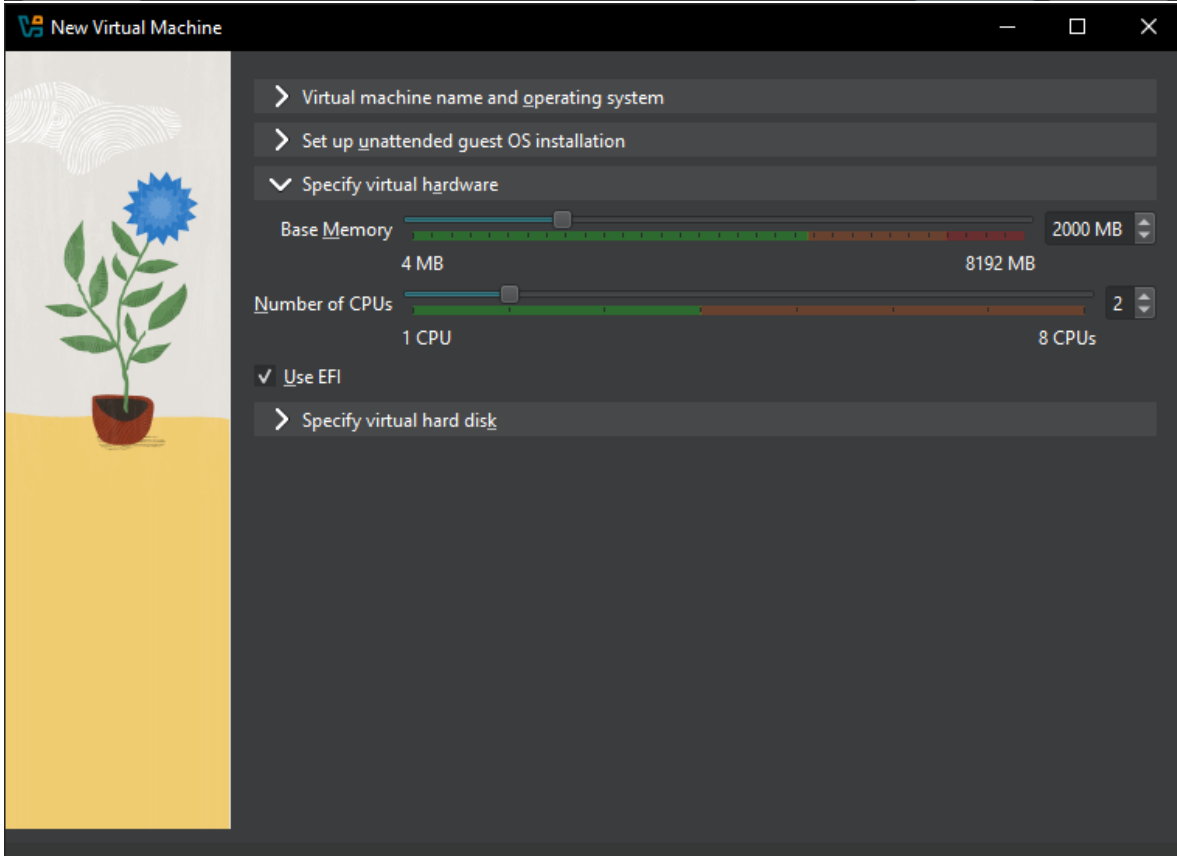
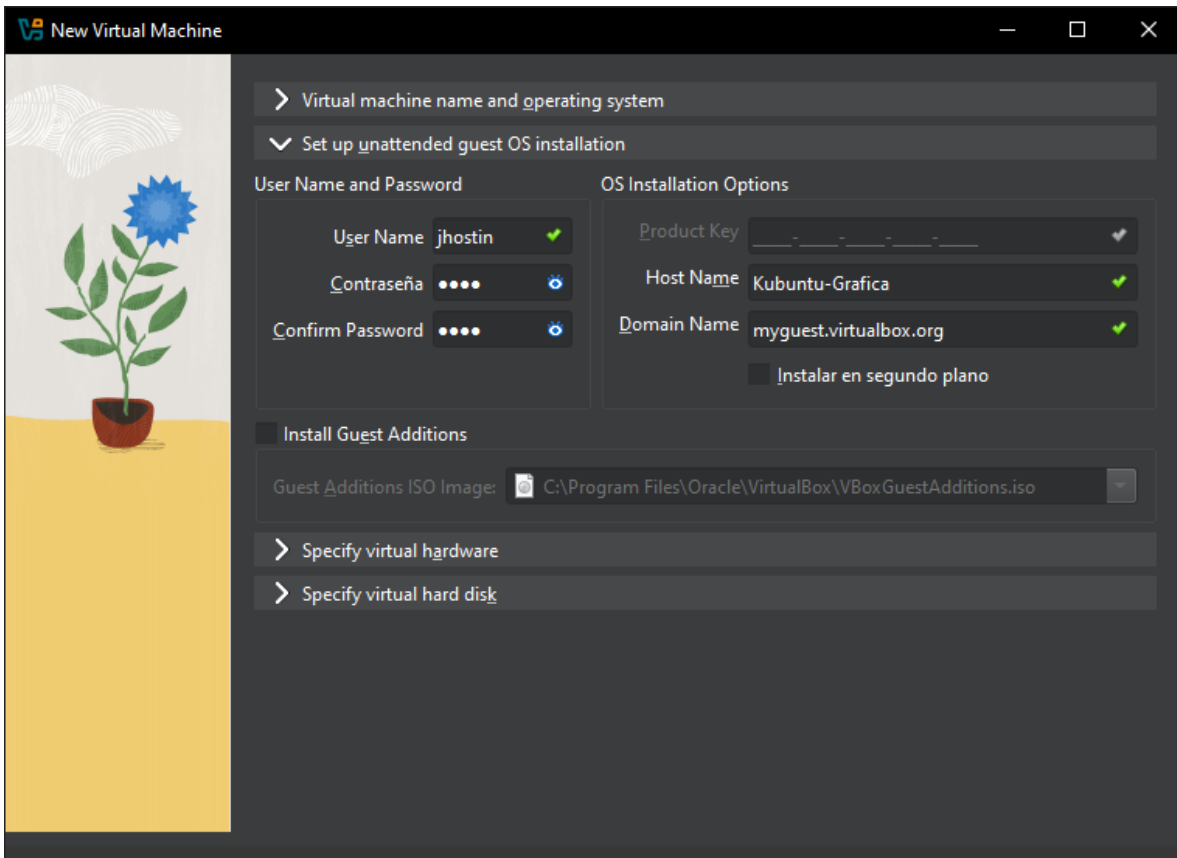


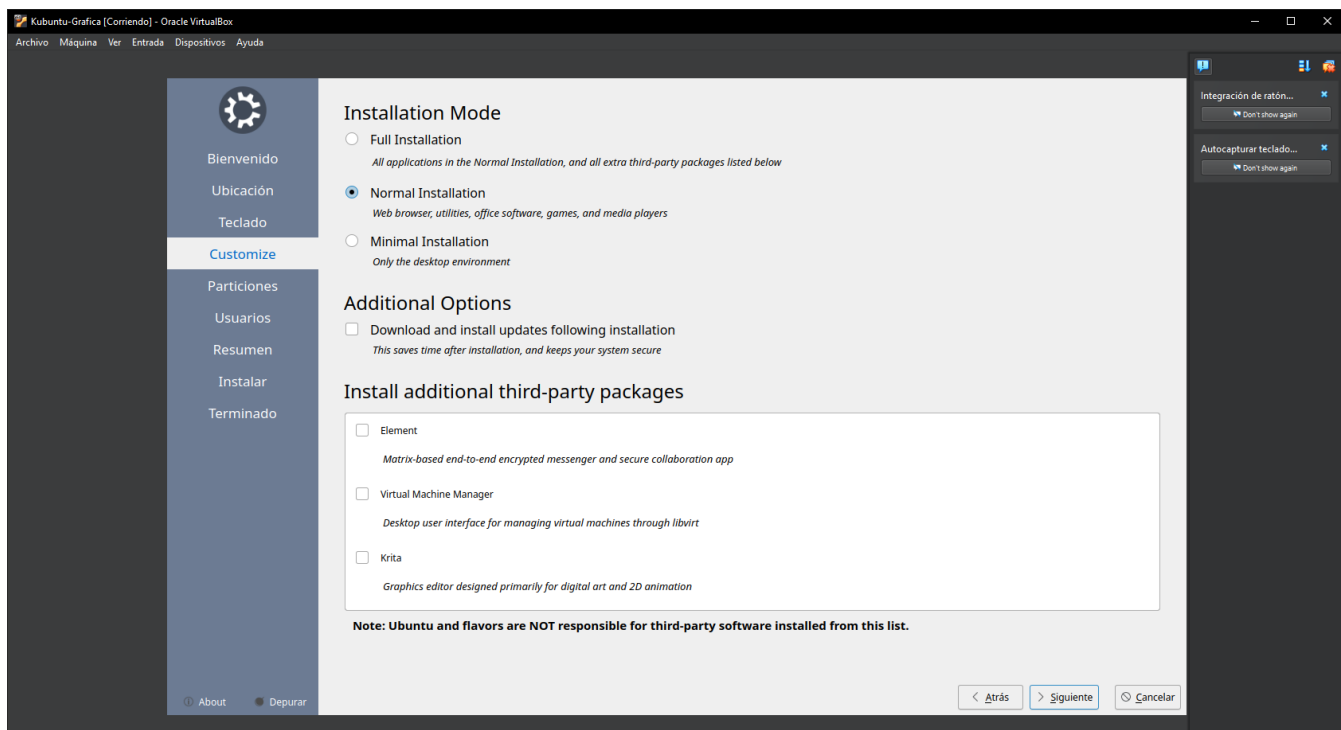
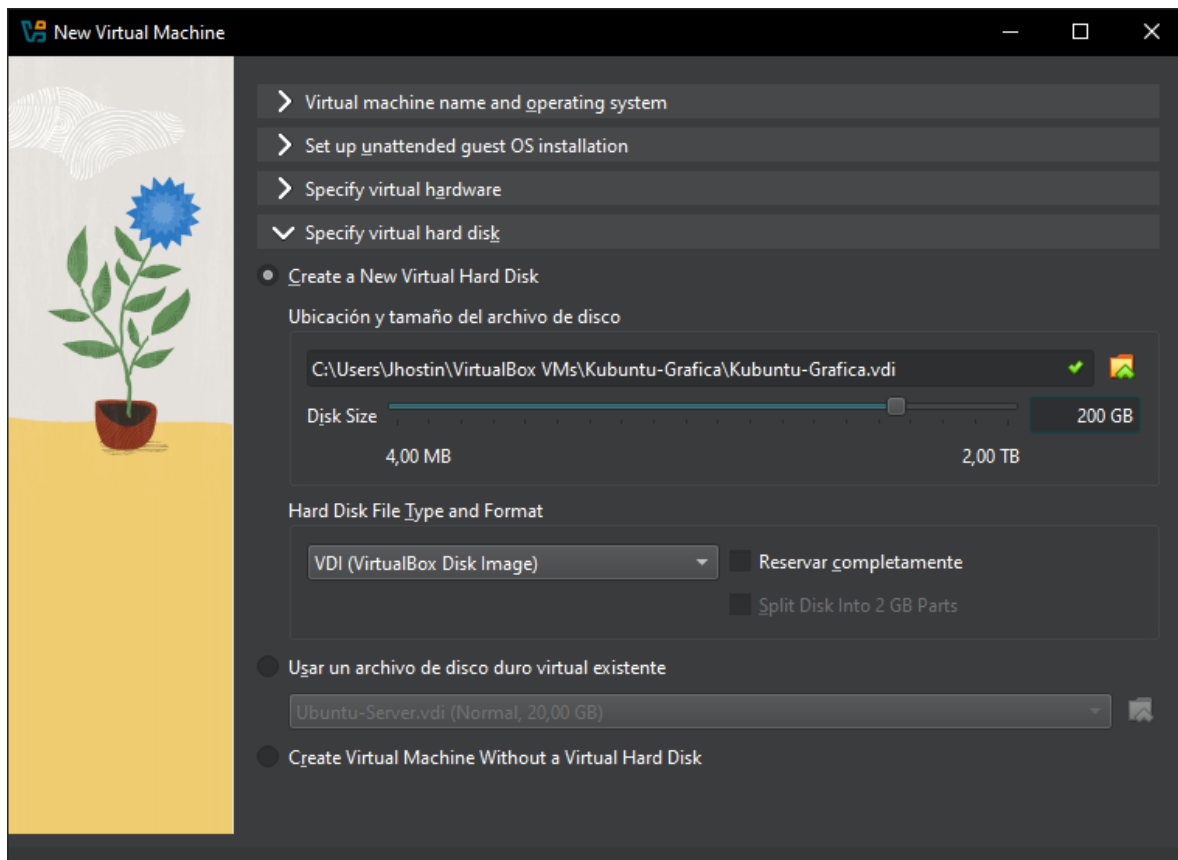
Incluir SSH server



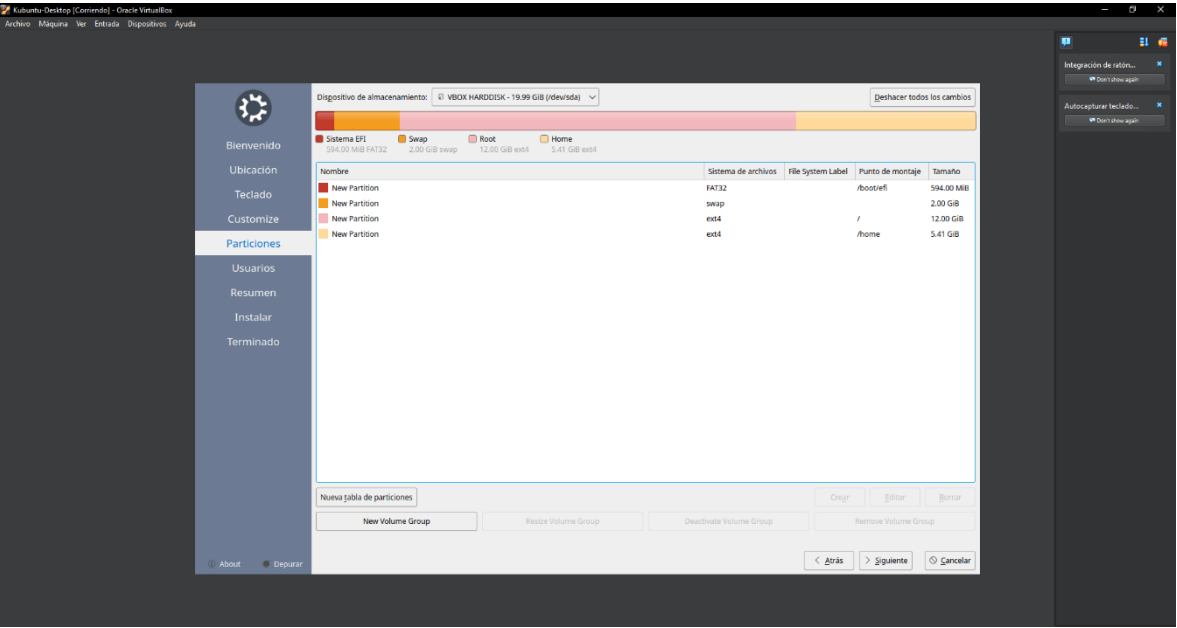
1.2 crear VM grafica



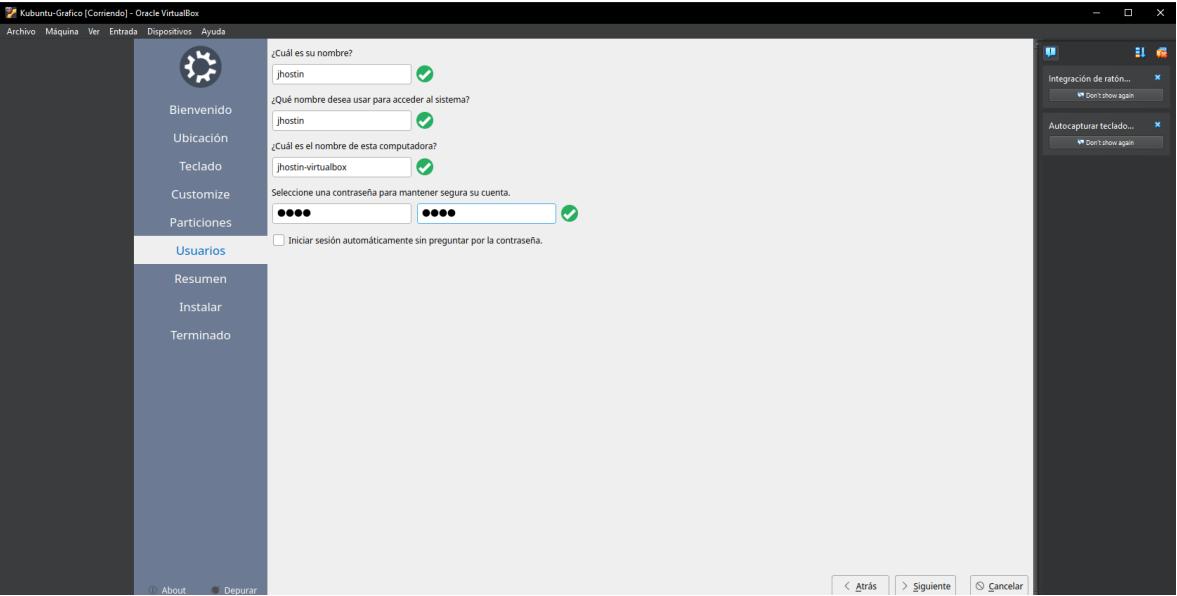




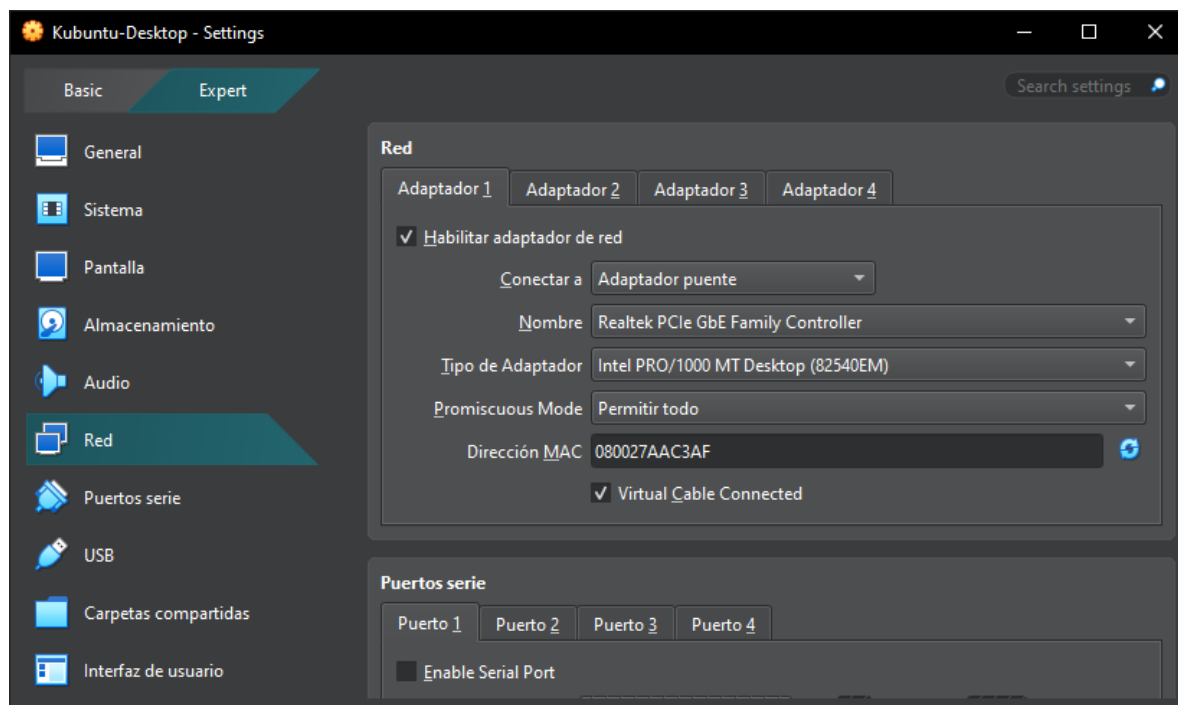
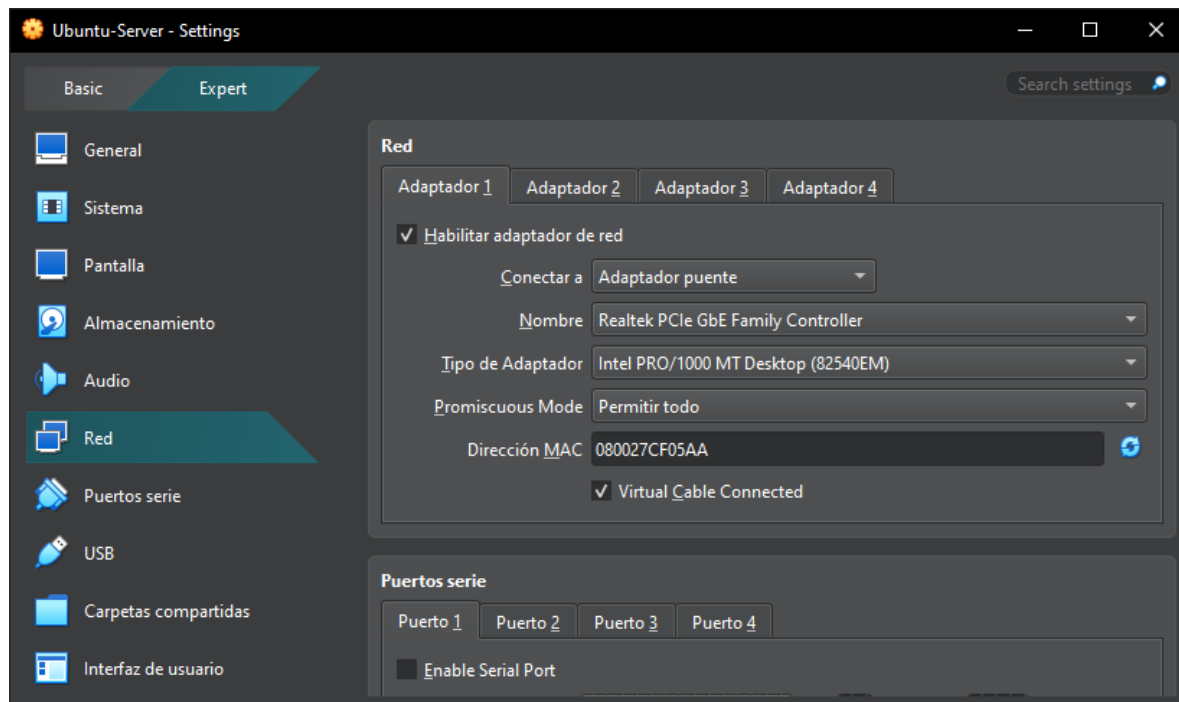
Creación de las particiones



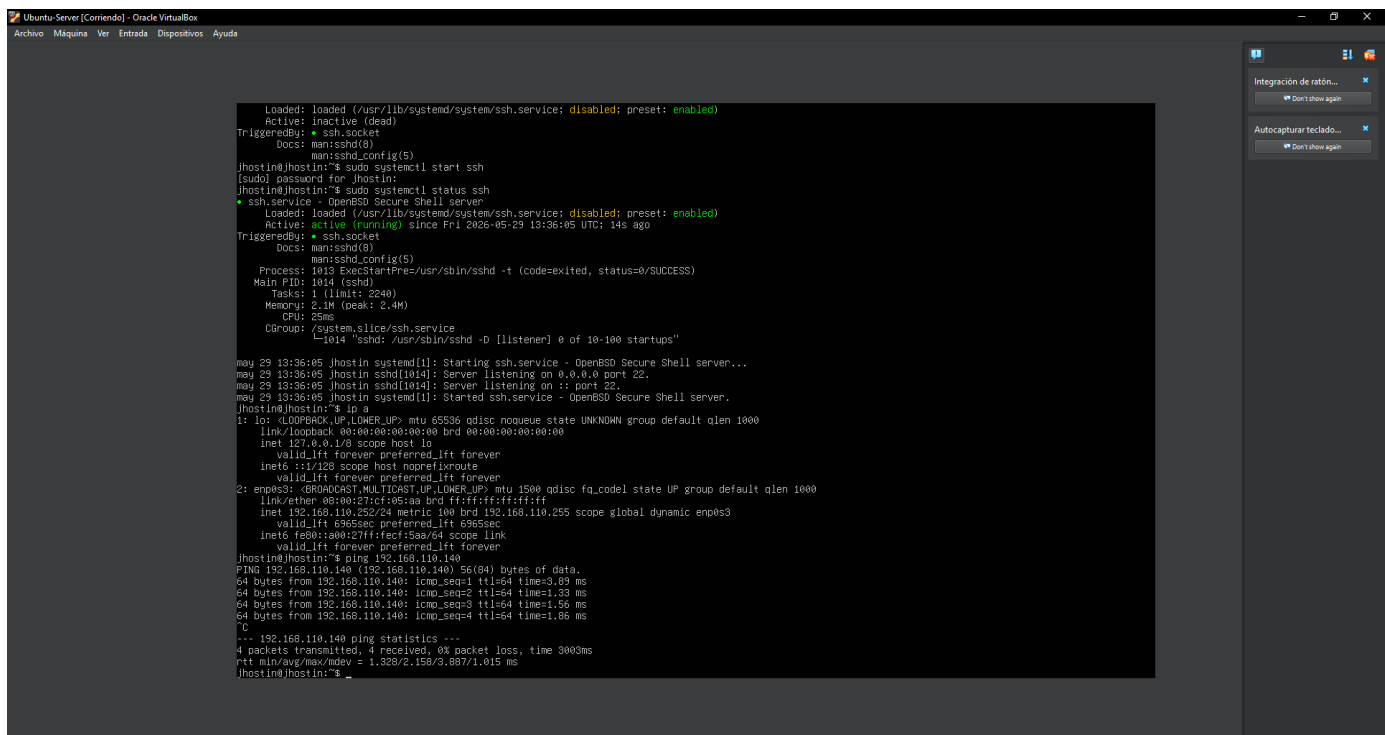
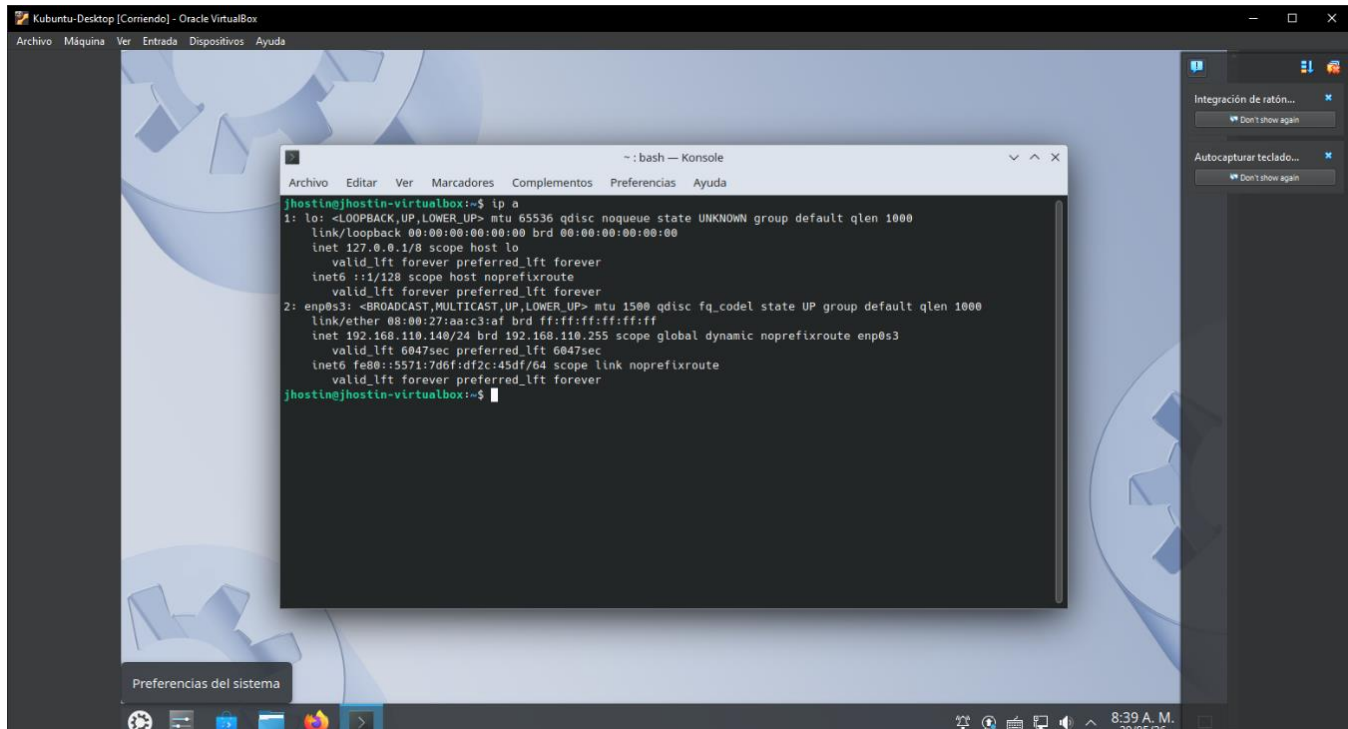
Creación del usuario



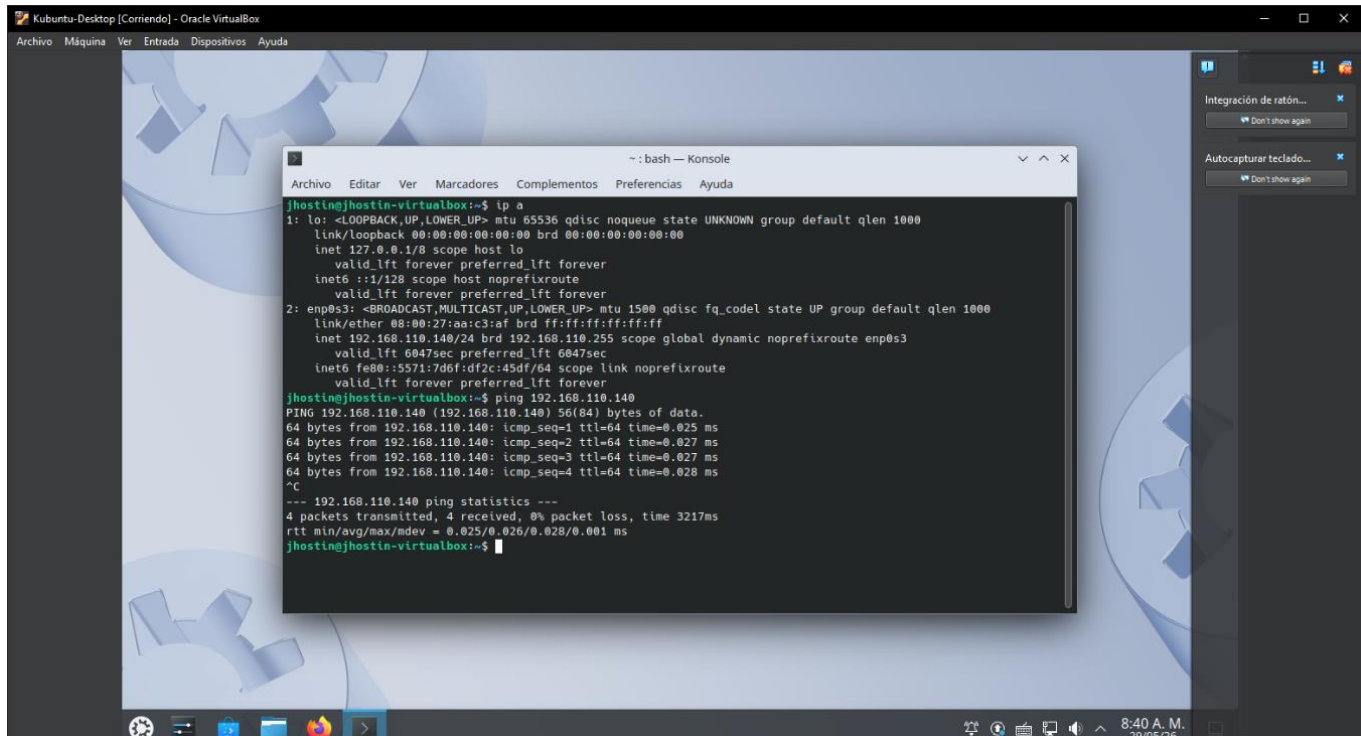
1.3 Configurar red interna entre ambas maquinas desde VirtualBox, en configuración de cada una se entra a red y se escoge el adaptador puente y promiscuous mode permitir todo



1.4 Verificar IP en cada una con el comando “ip a”

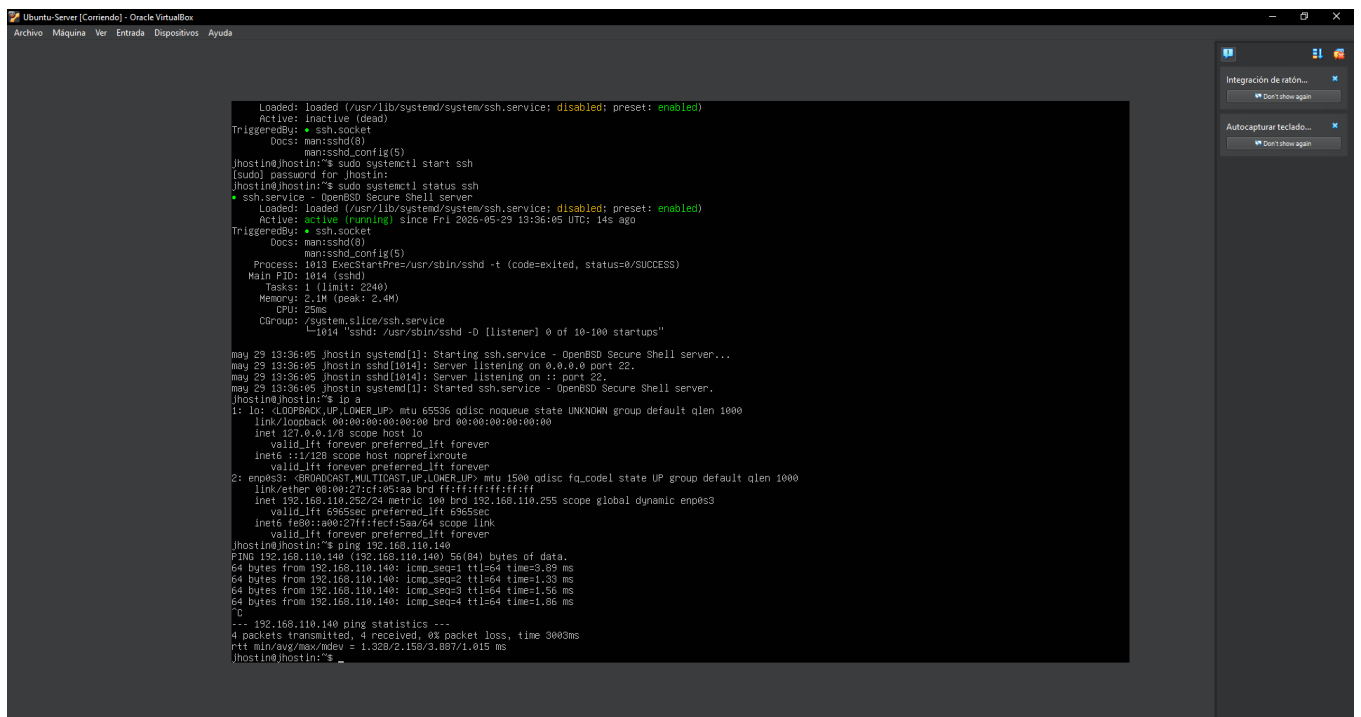


1.5 Confirmar que las maquinas se vean entre si con el comando “ping 192.168.110.140”



```
Kubuntu-Desktop [Corriendo] - Oracle VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda

jhostine@jhostin-virtualbox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:aa:c3:af brd ff:ff:ff:ff:ff:ff
    inet 192.168.110.140/24 brd 192.168.110.255 scope global dynamic noprefixroute enp0s3
        valid_lft 6047sec preferred_lft 6047sec
    inet6 fe80::5571:7d6f:df2c:45df/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
jhostine@jhostin-virtualbox:~$ ping 192.168.110.140
PING 192.168.110.140 (192.168.110.140) 56(84) bytes of data.
 64 bytes from 192.168.110.140: icmp_seq=1 ttl=64 time=0.025 ms
 64 bytes from 192.168.110.140: icmp_seq=2 ttl=64 time=0.027 ms
 64 bytes from 192.168.110.140: icmp_seq=3 ttl=64 time=0.027 ms
 64 bytes from 192.168.110.140: icmp_seq=4 ttl=64 time=0.028 ms
^C
--- 192.168.110.140 ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3217ms
 rtt min/avg/max/mdev = 0.025/0.026/0.028/0.001 ms
jhostine@jhostin-virtualbox:~$
```

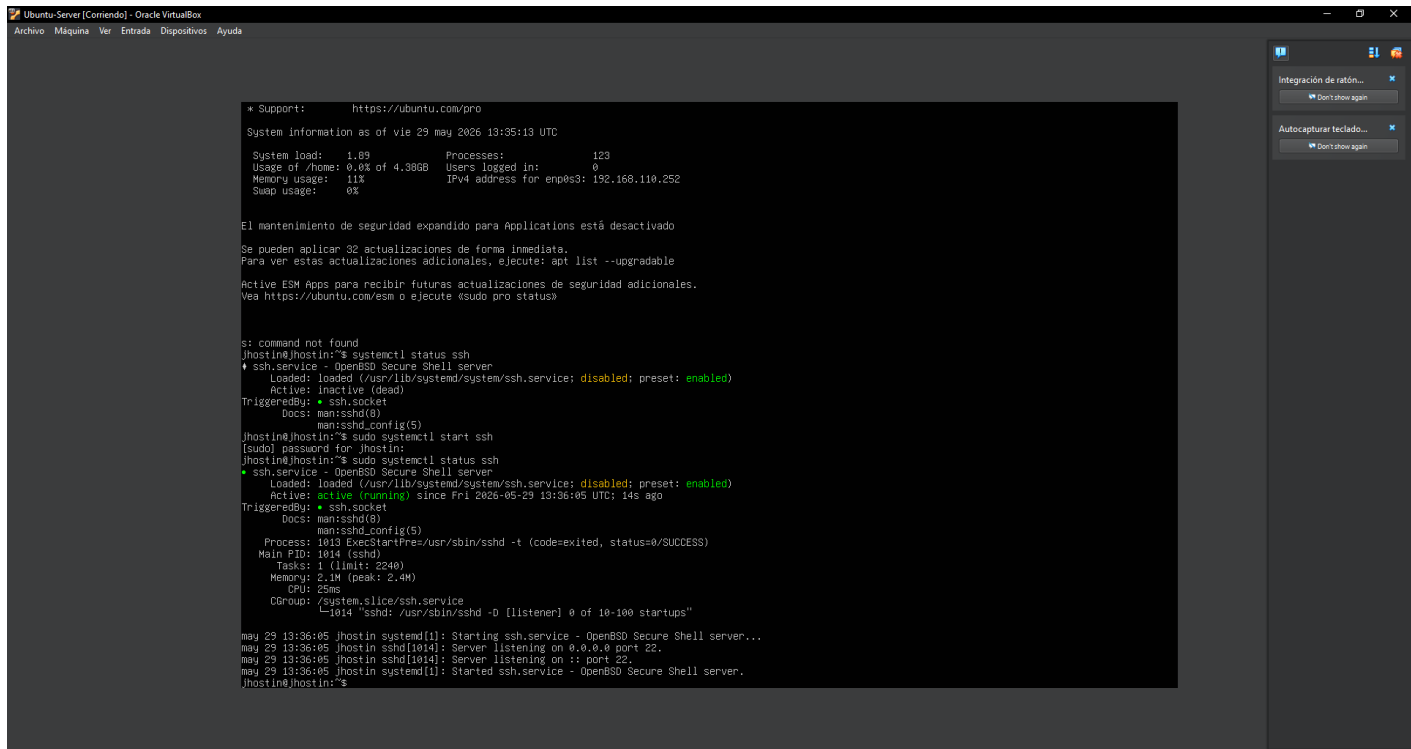


```
Ubuntu-Server [Corriendo] - Oracle VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda

Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
Active: inactive (dead)
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
      man:ssh_config(5)
jhostine@jhostin:~$ sudo systemctl start ssh
[sudo] password for jhostine:
jhostine@jhostin:~$ sudo systemctl status ssh
● ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2025-05-29 13:36:05 UTC; 14s ago
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
         man:ssh_config(5)
  Process: 1013 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 1014 (sshd)
       Tasks: 1 (limit: 2340)
      Memory: 2.1M (peak: 2.4M)
         CPU: 25ms
   CGroup: /system.slice/ssh.service
           └─1014 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

may 29 13:36:05 jhostin systemd[1]: Starting ssh.service - OpenSSH Secure Shell server...
may 29 13:36:05 jhostin sshd[1014]: Server listening on 0.0.0.0 port 22.
may 29 13:36:05 jhostin sshd[1014]: Server listening on :: port 22.
may 29 13:36:05 jhostin systemd[1]: Started ssh.service - OpenSSH Secure Shell server.
jhostine@jhostin:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cf:05:aa brd ff:ff:ff:ff:ff:ff
    inet 192.168.110.252/24 metric 100 brd 192.168.110.255 scope global dynamic enp0s3
        valid_lft 595sec preferred_lft 595sec
    inet6 fe80::a00:27ff:fecl:5aa/64 scope link
        valid_lft forever preferred_lft forever
jhostine@jhostin:~$ ping 192.168.110.140
PING 192.168.110.140 (192.168.110.140) 56(84) bytes of data.
 64 bytes from 192.168.110.140: icmp_seq=1 ttl=64 time=3.89 ms
 64 bytes from 192.168.110.140: icmp_seq=2 ttl=64 time=1.33 ms
 64 bytes from 192.168.110.140: icmp_seq=3 ttl=64 time=1.56 ms
 64 bytes from 192.168.110.140: icmp_seq=4 ttl=64 time=1.06 ms
^C
--- 192.168.110.140 ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3003ms
 rtt min/avg/max/mdev = 1.328/2.158/3.887/1.015 ms
jhostine@jhostin:~$
```

1.6 Configurar SSH: con “systemctl status ssh” verifica que esta inactiva por lo que se usa “systemctl start ssh” para activarla



```
* Support: https://ubuntu.com/pro

System information as of vie 29 may 2026 13:35:13 UTC

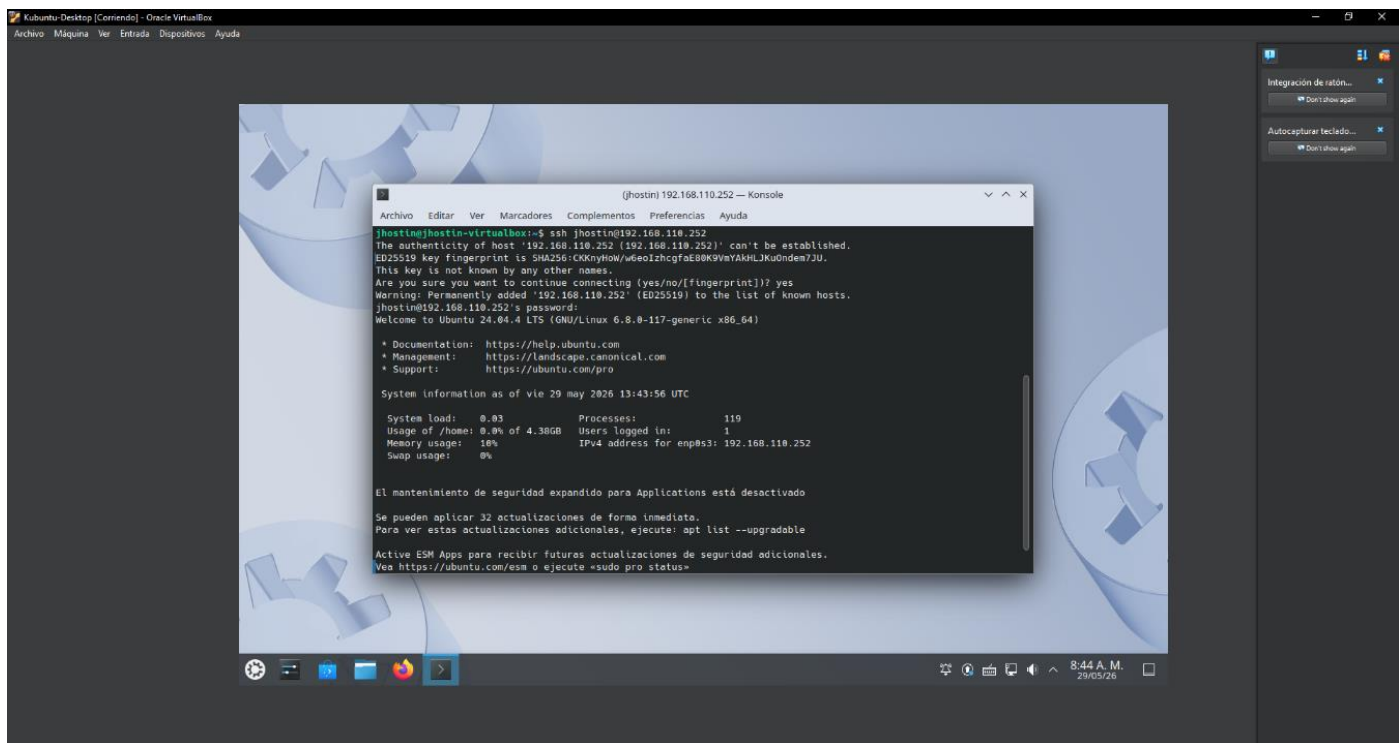
System load: 1.89 Processes: 123
Usage of /home: 0.0% of 4.38GB Users logged in: 0
Memory usage: 11% IPv4 address for enp0s3: 192.168.110.252
Swap usage: 0%

El mantenimiento de seguridad expandido para Applications está desactivado
Se pueden aplicar 32 actualizaciones de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable
Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

si: command not found
jhostin@jhostin:~$ systemctl status ssh
* ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:ssh(8)
         man:ssh_config(5)
jhostin@jhostin:~$ sudo systemctl start ssh
[sudo] password for jhostin:
jhostin@jhostin:~$ sudo systemctl status ssh
* ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-05-29 13:36:05 UTC; 14s ago
   TriggeredBy: ● ssh.socket
   Docs: man:ssh(8)
         man:ssh_config(5)
   Process: 1013 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1014 (sshd)
   Tasks: 1 (limit: 2440)
   Memory: 2.1M (peak: 2.4M)
   CPU: 25ms
   CGroup: /system.slice/ssh.service
           └─1014 "sshd: /usr/sbin/sshd -O [listener] 0 of 10-100 startups"

may 29 13:36:05 jhostin systemd[1]: Starting ssh.service - OpenSSH Secure Shell server...
may 29 13:36:05 jhostin sshd[1014]: Server listening on 0.0.0.0 port 22.
may 29 13:36:05 jhostin sshd[1014]: Server listening on :: port 22.
may 29 13:36:05 jhostin systemd[1]: Started ssh.service - OpenSSH Secure Shell server.
jhostin@jhostin:~$
```

1.7 En desktop usar el comando ssh [jhostin@198.168.110.252](https://198.168.110.252) para conectarse con el servidor



```
(jhostin) 192.168.110.252 — Konsole
Archivo Editar Ver Marcadores Complementos Preferencias Ayuda

jhostin@jhostin-virtualbox:~$ ssh jhostin@192.168.110.252
The authenticity of host '192.168.110.252 (192.168.110.252)' can't be established.
ED25519 key fingerprint is SHA256:CKKnyHow/w6oIzhcgfaE80K9VnYAKHLJKuonden7JU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.110.252' (ED25519) to the list of known hosts.
jhostin@192.168.110.252's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-117-generic x86_64)

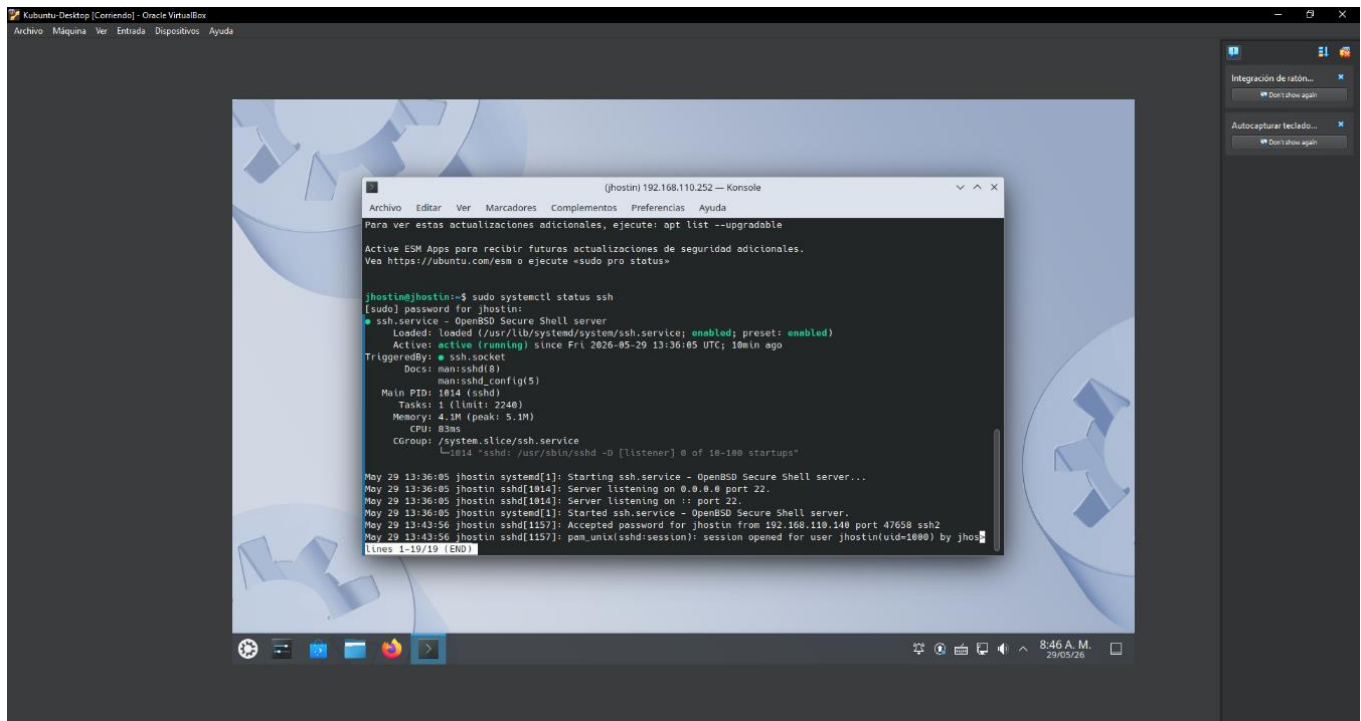
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of vie 29 may 2026 13:43:56 UTC

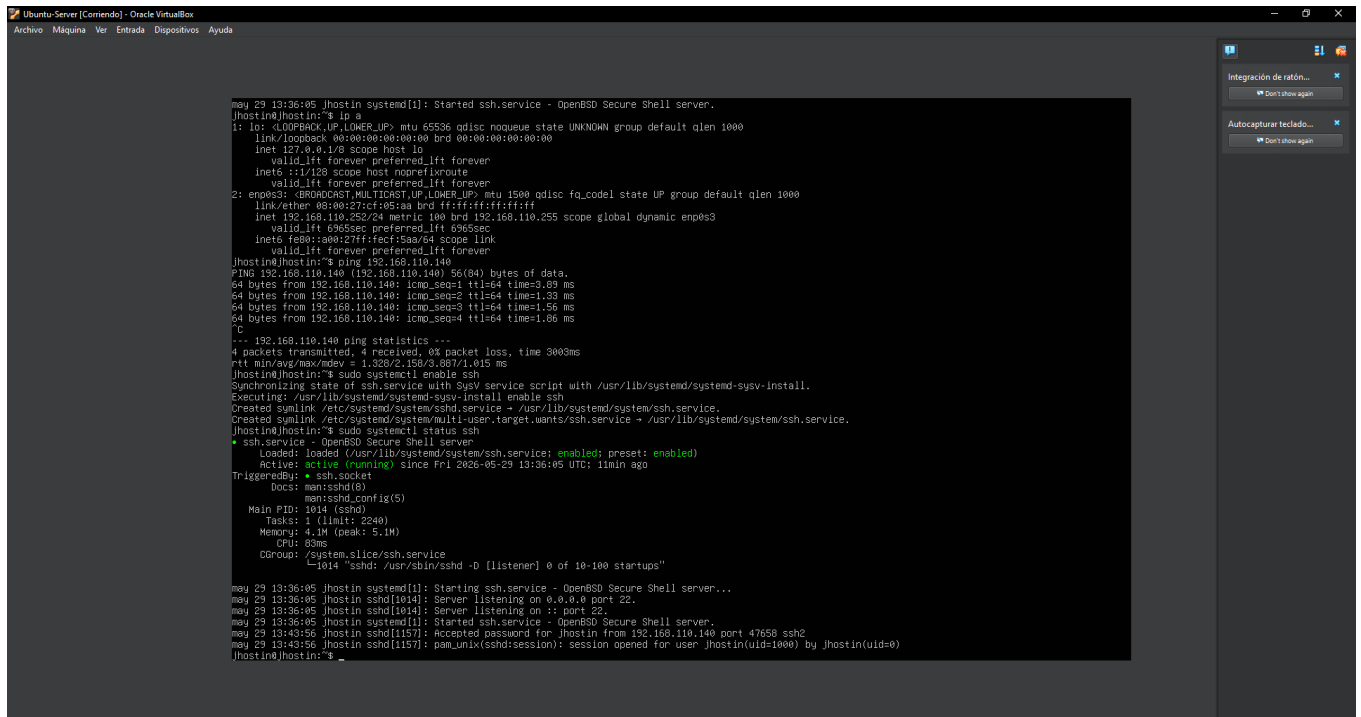
System load: 0.03 Processes: 119
Usage of /home: 0.0% of 4.38GB Users logged in: 1
Memory usage: 18% IPv4 address for enp0s3: 192.168.110.252
Swap usage: 0%

El mantenimiento de seguridad expandido para Applications está desactivado
Se pueden aplicar 32 actualizaciones de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable
Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»
```

Se confirma la conexión con el comando “sudo systemctl status ssh” desde Kubuntu desktop y Ubuntu server



The screenshot shows a Kubuntu Desktop environment with a terminal window titled "(jhostin) 192.168.110.252 — Konsole". The terminal displays the command `sudo systemctl status ssh` and its output. The output indicates that the `ssh.service` is loaded, active (running), and enabled. It also shows the service's configuration, including the main PID (1014), tasks (1), memory (4.1M), and CPU (33ms). The service is part of the `system.slice/ssh.service` group. The terminal also shows the service's status as "loaded" and "active (running)" since Fri 2020-05-29 13:36:05 UTC, 10min ago. The service is triggered by `ssh.socket` and `man:ssh(8)`. The main PID is 1014 (sshd), and it has 1 task (limit: 2240). The memory usage is 4.1M (peak: 5.1M) and the CPU usage is 33ms. The service is part of the `system.slice/ssh.service` group. The terminal also shows the service's status as "loaded" and "active (running)" since Fri 2020-05-29 13:36:05 UTC, 10min ago. The service is triggered by `ssh.socket` and `man:ssh(8)`. The main PID is 1014 (sshd), and it has 1 task (limit: 2240). The memory usage is 4.1M (peak: 5.1M) and the CPU usage is 33ms. The service is part of the `system.slice/ssh.service` group. The terminal also shows the service's status as "loaded" and "active (running)" since Fri 2020-05-29 13:36:05 UTC, 10min ago. The service is triggered by `ssh.socket` and `man:ssh(8)`. The main PID is 1014 (sshd), and it has 1 task (limit: 2240). The memory usage is 4.1M (peak: 5.1M) and the CPU usage is 33ms. The service is part of the `system.slice/ssh.service` group.



The screenshot shows an Ubuntu Server environment with a terminal window titled "Ubuntu Server [Comando] - Oracle VM VirtualBox". The terminal displays the command `sudo systemctl status ssh` and its output. The output indicates that the `ssh.service` is loaded, active (running), and enabled. It also shows the service's configuration, including the main PID (1014), tasks (1), memory (4.1M), and CPU (33ms). The service is part of the `system.slice/ssh.service` group. The terminal also shows the service's status as "loaded" and "active (running)" since Fri 2020-05-29 13:36:05 UTC, 11min ago. The service is triggered by `ssh.socket` and `man:ssh(8)`. The main PID is 1014 (sshd), and it has 1 task (limit: 2240). The memory usage is 4.1M (peak: 5.1M) and the CPU usage is 33ms. The service is part of the `system.slice/ssh.service` group. The terminal also shows the service's status as "loaded" and "active (running)" since Fri 2020-05-29 13:36:05 UTC, 11min ago. The service is triggered by `ssh.socket` and `man:ssh(8)`. The main PID is 1014 (sshd), and it has 1 task (limit: 2240). The memory usage is 4.1M (peak: 5.1M) and the CPU usage is 33ms. The service is part of the `system.slice/ssh.service` group.

1.8 verificar las particiones en cada VM con lsblk

```
Ubuntu-Server [Comando] - Oracle VM VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda

Ubuntu 24.04.4 LTS jhostin tty1
jhostin login: jhostin
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-117-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of vie 29 may 2026 14:12:35 UTC

System load:  1.1          Processes:      120
Usage of /home: 6.0% of 4.8GB    Users logged in: 0
Memory usage:  9%          IPv4 address for enp0s3: 192.168.110.252
Swap usage:    0%

El mantenimiento de seguridad expandido para Applications está desactivado
Se pueden aplicar 32 actualizaciones de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable
Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

jhostin@jhostin:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0   256G  0 disk
├─sda1      8:1    0   953M  0 part /boot/efi
├─sda2      8:2    0    12G  0 part /
├─sda3      8:3    0    553M  0 part 
├─sda4      8:4    0     2G  0 part [SWAP]
├─sda5      8:5    0   4.5G  0 part /home
└─sda6     11:0    1  1802M  0 rom

jhostin@jhostin:~$
```

```
Kubuntu-Desktop [Comando] - Oracle VM VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda

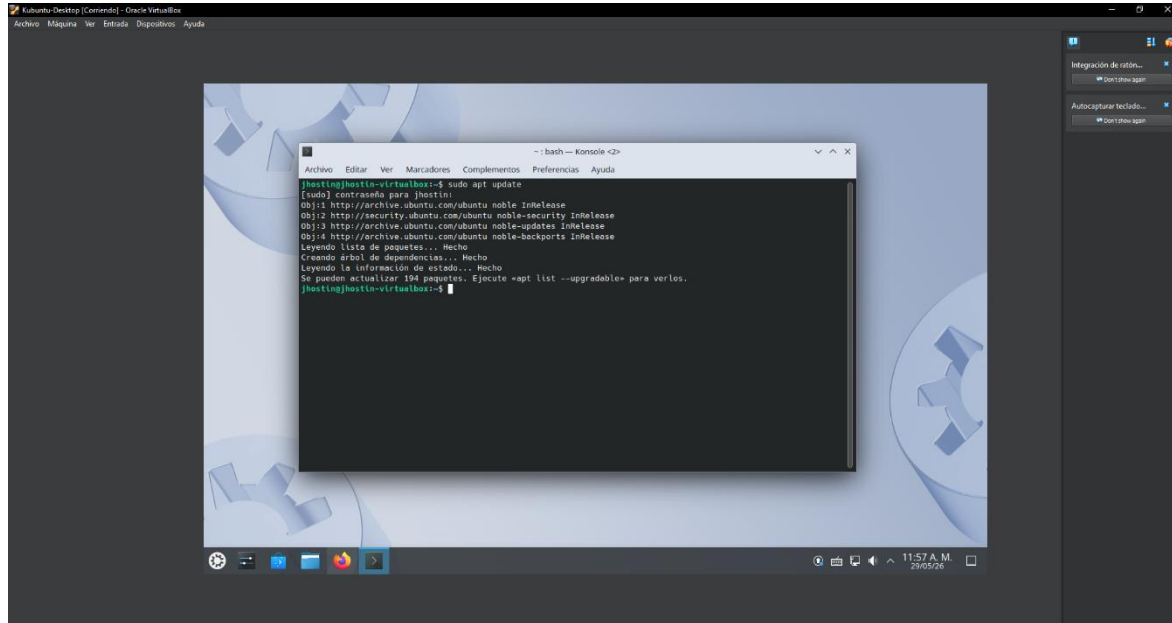
~: bash -- Konsole
Archivo Editar Ver Marcadores Complementos Preferencias Ayuda

jhostin@jhostin-virtualbox:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0     4K  1 loop /snap/bare/5
loop1       7:1    0    74M  1 loop /snap/core22/2292
loop2       7:2    0   251.7M  1 loop /snap/firefox/7766
loop3       7:3    0   531.6M  1 loop /snap/gnome-15-2204/247
loop4       7:4    0   18.5M  1 loop /snap/fireware-updater/210
loop5       7:5    0   48.1M  1 loop /snap/snapd/25935
loop6       7:6    0   91.7M  1 loop /snap/gtk-common-themes/1535
loop7       7:7    0  226.6M  1 loop /snap/thunderbird/959
sda         8:0    0   256G  0 disk
├─sda1      8:1    0   594M  0 part /boot/efi
├─sda2      8:2    0    12G  0 part /
├─sda3      8:3    0     2G  0 part [SWAP]
├─sda4      8:4    0   5.4G  0 part /home
└─sda5     11:0    1  1824M  0 rom

jhostin@jhostin-virtualbox:~$
```

2.Contenedores Docker

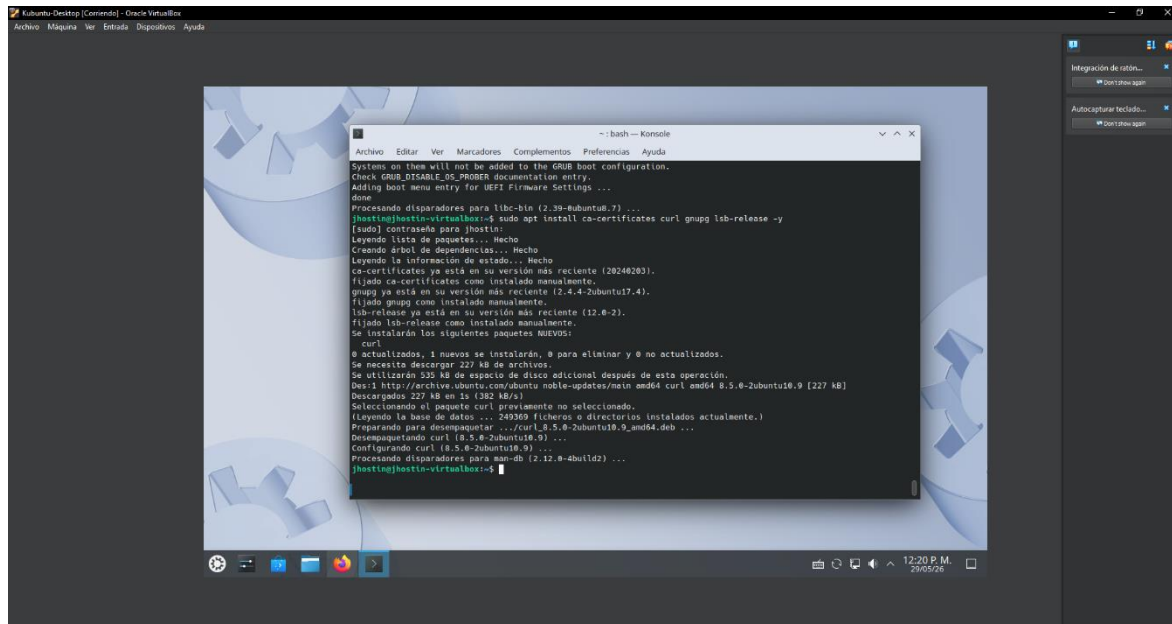
2.1 actualizar sistema con “sudo apt update”



```
--: bash -- Konsole -x2-
Archivo Editar Ver Marcadores Complementos Preferencias Ayuda

jhosting@hostin-virtualbox:~$ sudo apt update
[sudo] contraseña para jhostin:
Obj:1 http://archive.ubuntu.com/ubuntu noble InRelease
Obj:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Obj:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Obj:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se pueden actualizar 194 paquetes. Ejecute «apt list --upgradable» para verlos.
jhosting@hostin-virtualbox:~$
```

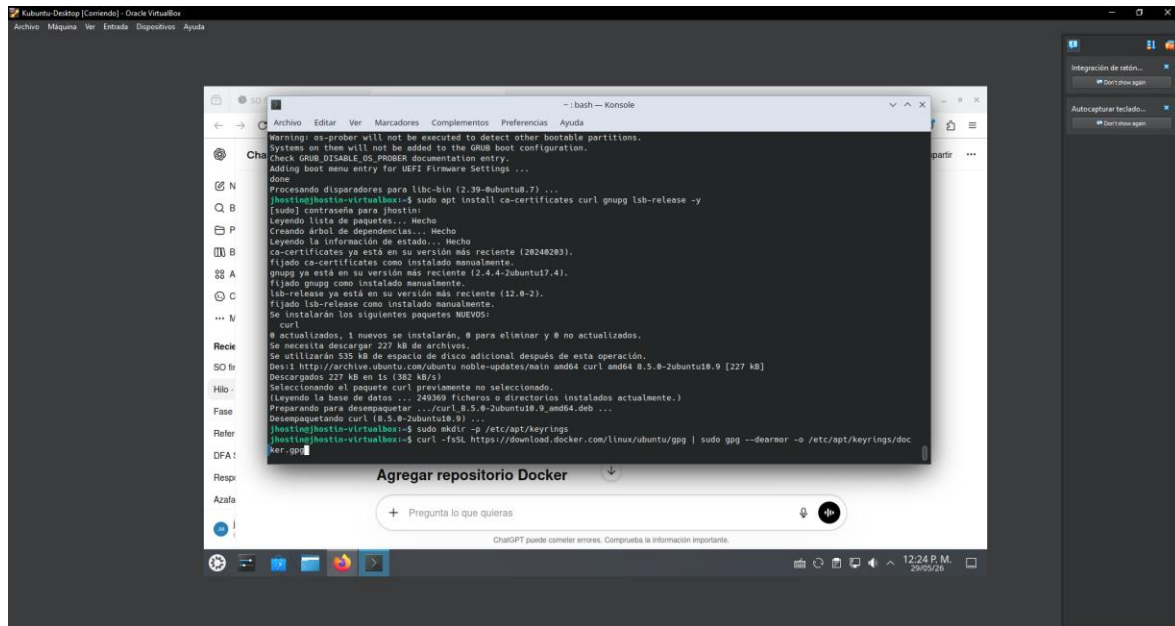
2.2 instalar dependencias con “sudo apt install ca-certificates curl gnupg lsb-release -y”



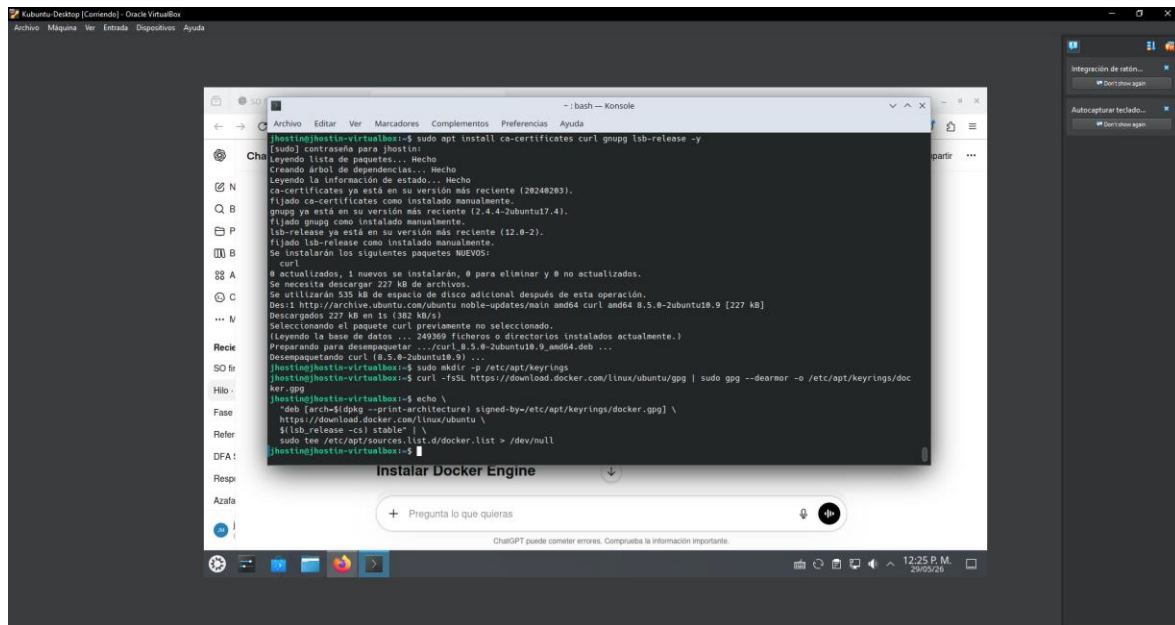
```
--: bash -- Konsole
Archivo Editar Ver Marcadores Complementos Preferencias Ayuda

Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Procesando disparadores para libc-bin (2.39-0ubuntu7) ...
jhosting@hostin-virtualbox:~$ sudo apt install ca-certificates curl gnupg lsb-release -y
[sudo] contraseña para jhostin:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
ca-certificates ya está en su versión más reciente (20240203).
Fijado ca-certificates como instalado manualmente.
gnupg ya está en su versión más reciente (2.4.4-2ubuntu17.4).
Fijado gnupg como instalado manualmente.
lsb-release ya está en su versión más reciente (12.0-2).
Fijado lsb-release como instalado manualmente.
Se instalarán los siguientes paquetes nuevos:
  curl
0 actualizados, 1 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
Se necesitan descargar 227 kB de archivos.
Se utilizarán 535 kB de espacio de disco adicional después de esta operación.
Des1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.9 (227 kB)
Descargados 227 kB en 1s (352 kB/s)
Seleccionando el paquete curl previamente no seleccionado.
(Leyendo la base de datos ... 249369 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../curl.8.5.0-2ubuntu10.9_amd64.deb ...
Desempaquetando curl (8.5.0-2ubuntu10.9) ...
Configurando curl (8.5.0-2ubuntu10.9) ...
Procesando disparadores para man-db (2.12.0-4build2) ...
jhosting@hostin-virtualbox:~$
```

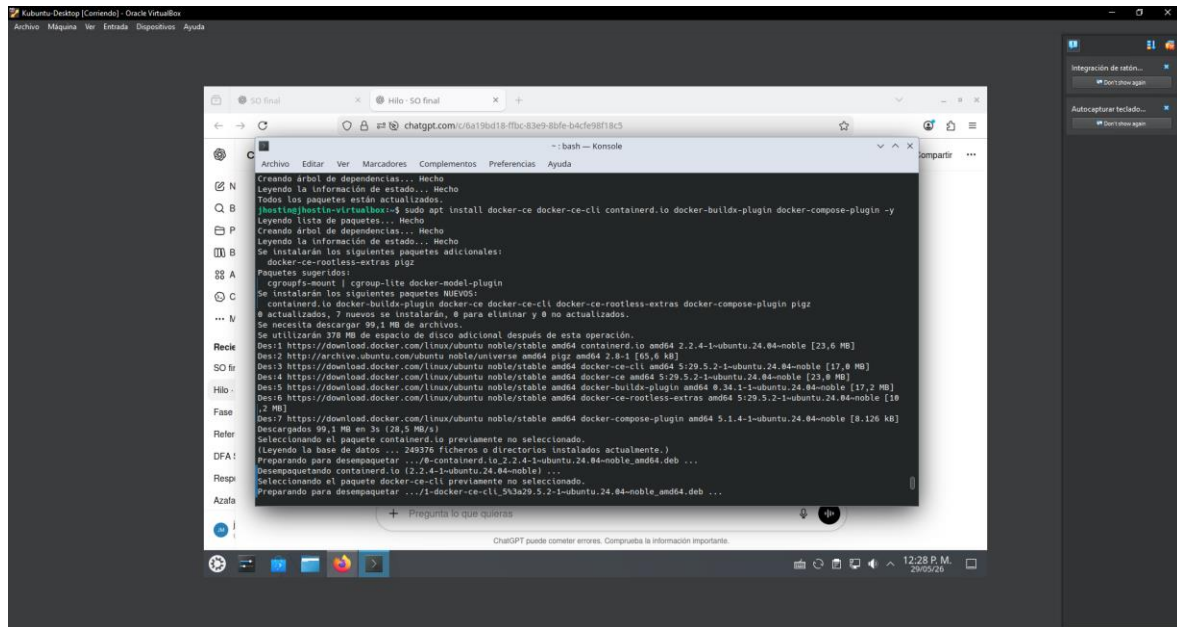
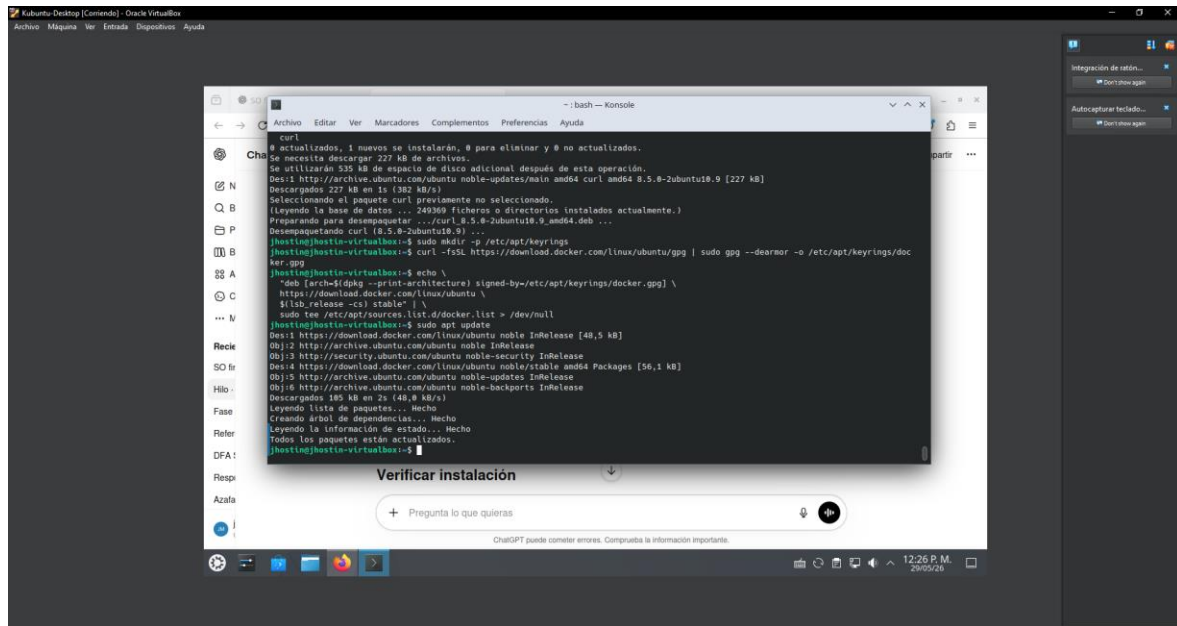
2.3 agregar llave oficial Docker con “curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg”



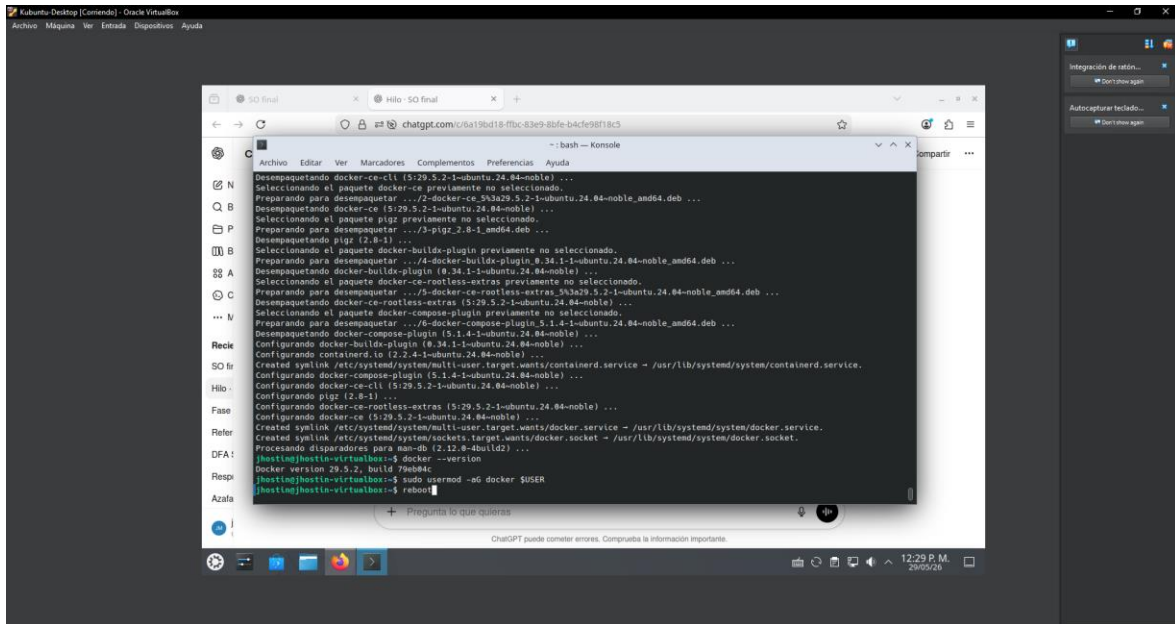
2.4 agregar repositorio Docker con “echo \n\"deb [arch=\$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \nhttps://download.docker.com/linux/ubuntu \n\$(. /etc/os-release && echo \"\$VERSION_CODENAME\") stable\" | \nsudo tee /etc/apt/sources.list.d/docker.list > /dev/null”



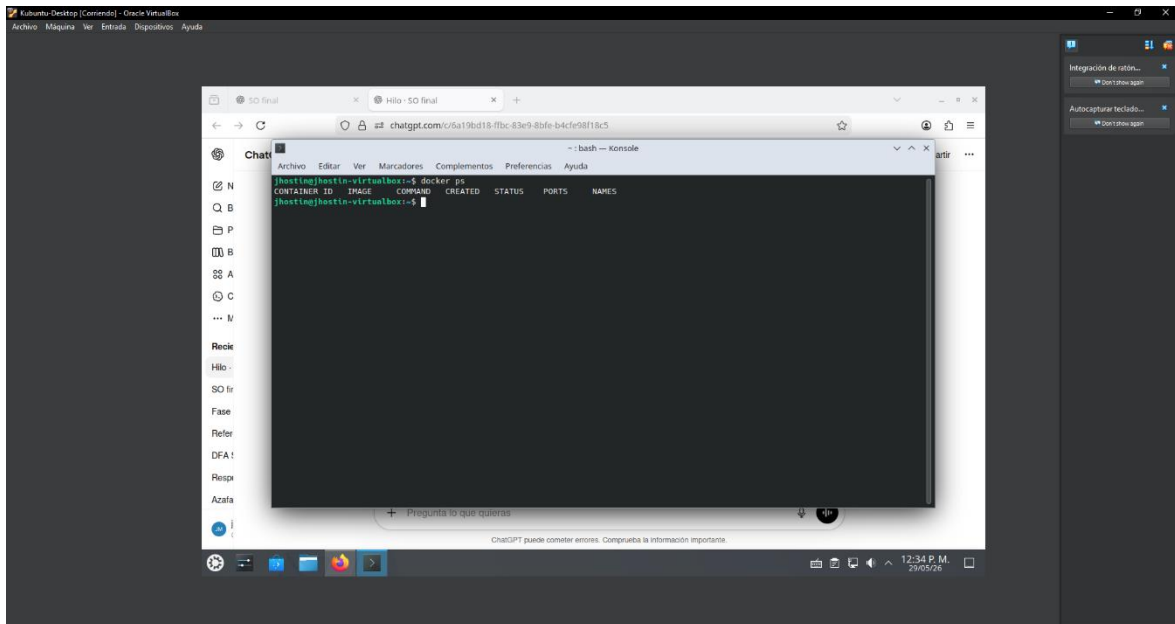
2.5 instalar Docker engine con “sudo apt update” y “sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin”



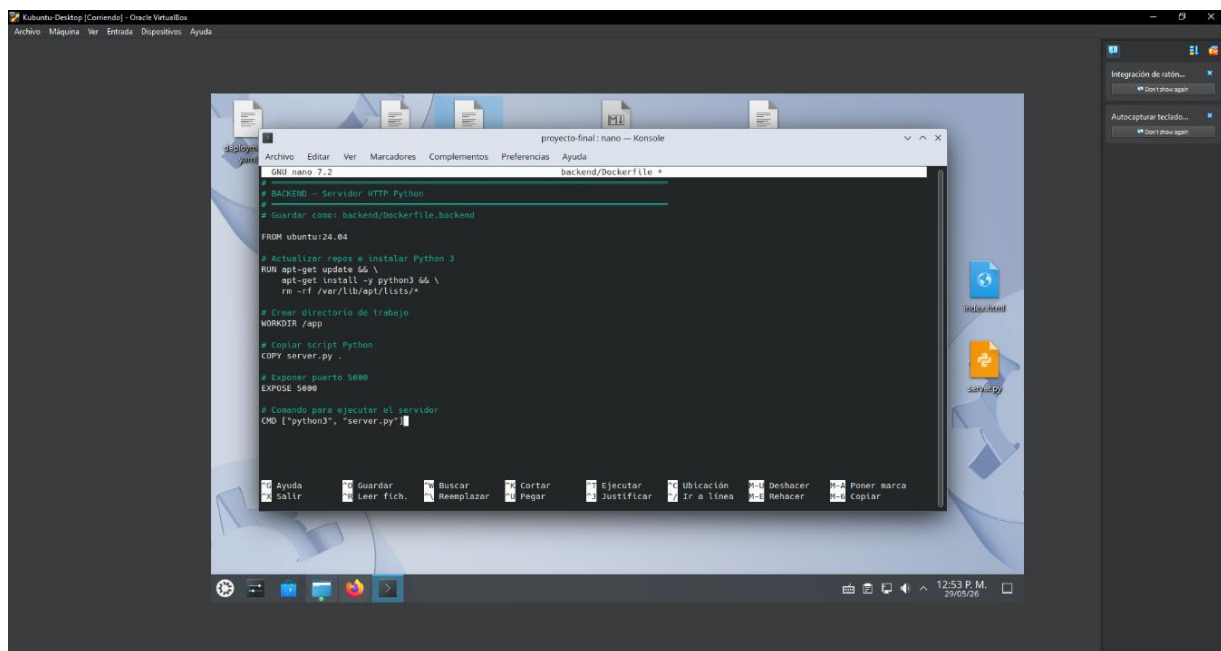
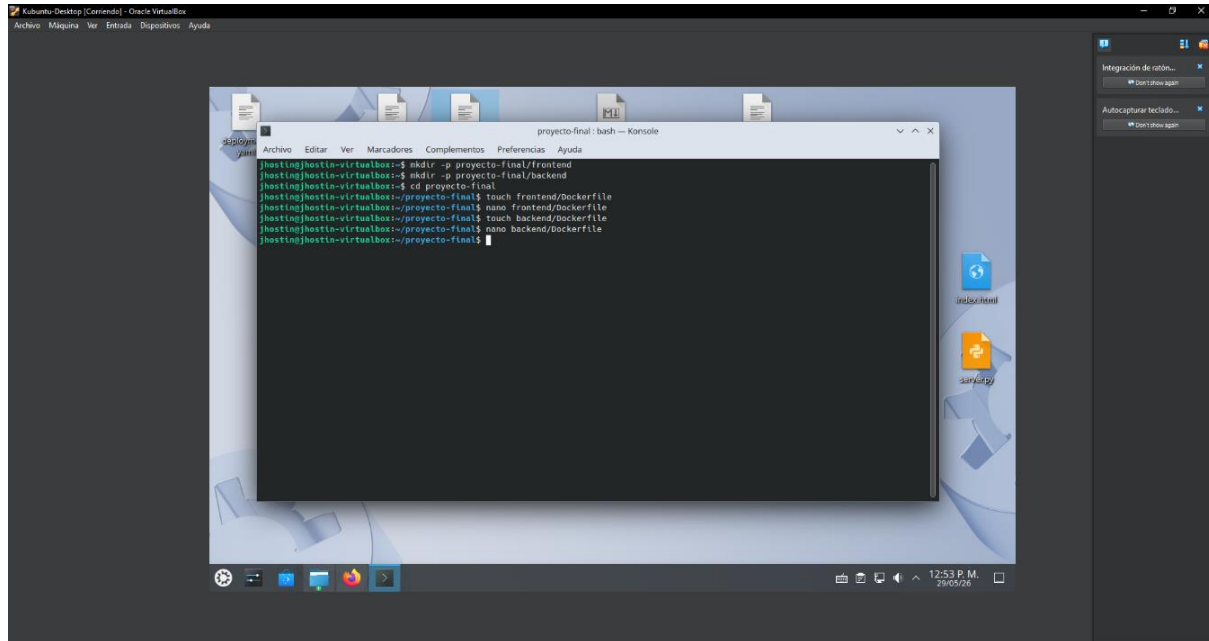
2.6 verificar instalación “docker --version”, permitir Docker sin sudo “sudo usermod -aG Docker \$USER” y reiniciar para aplicar cambios con “reboot”

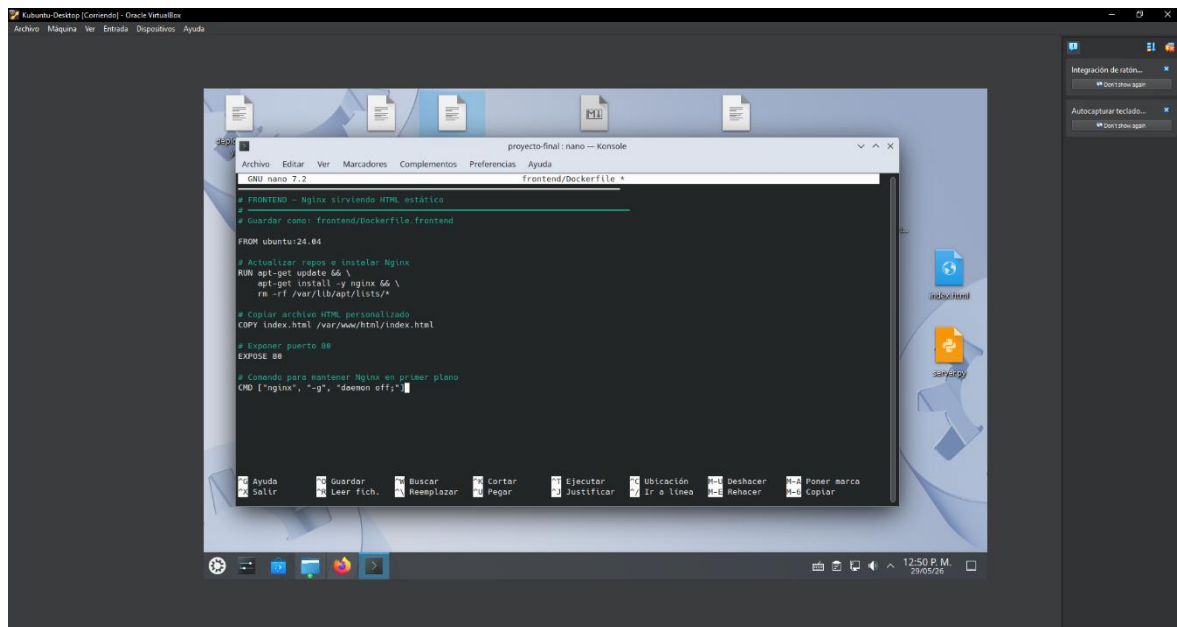


2.7 verificar Docker con “docker ps”

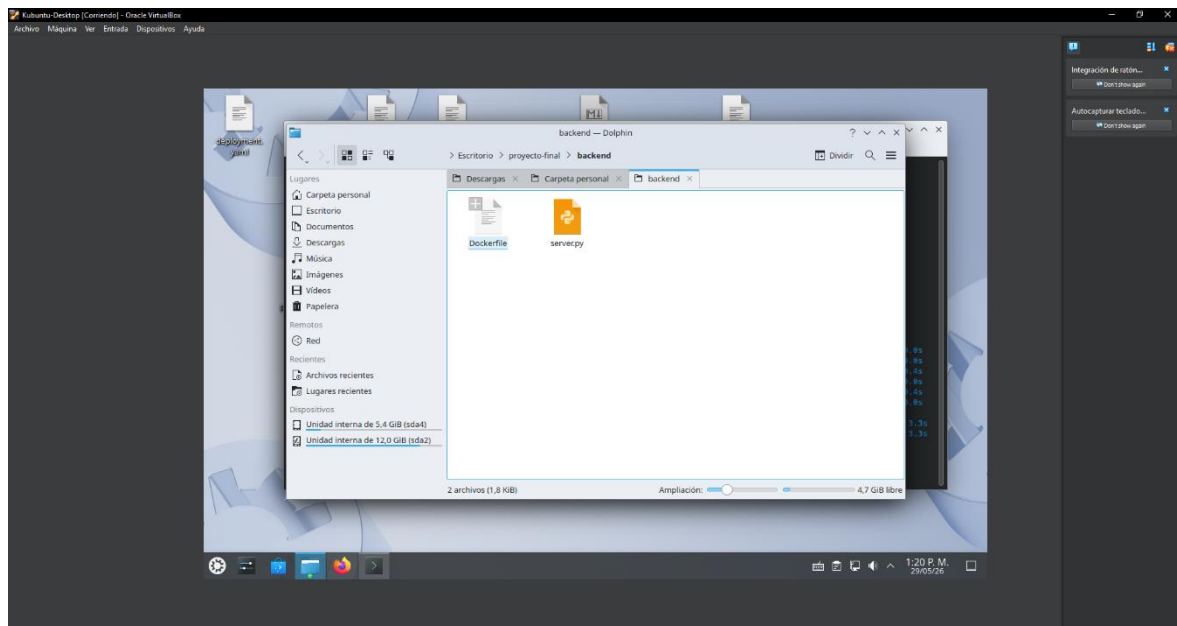
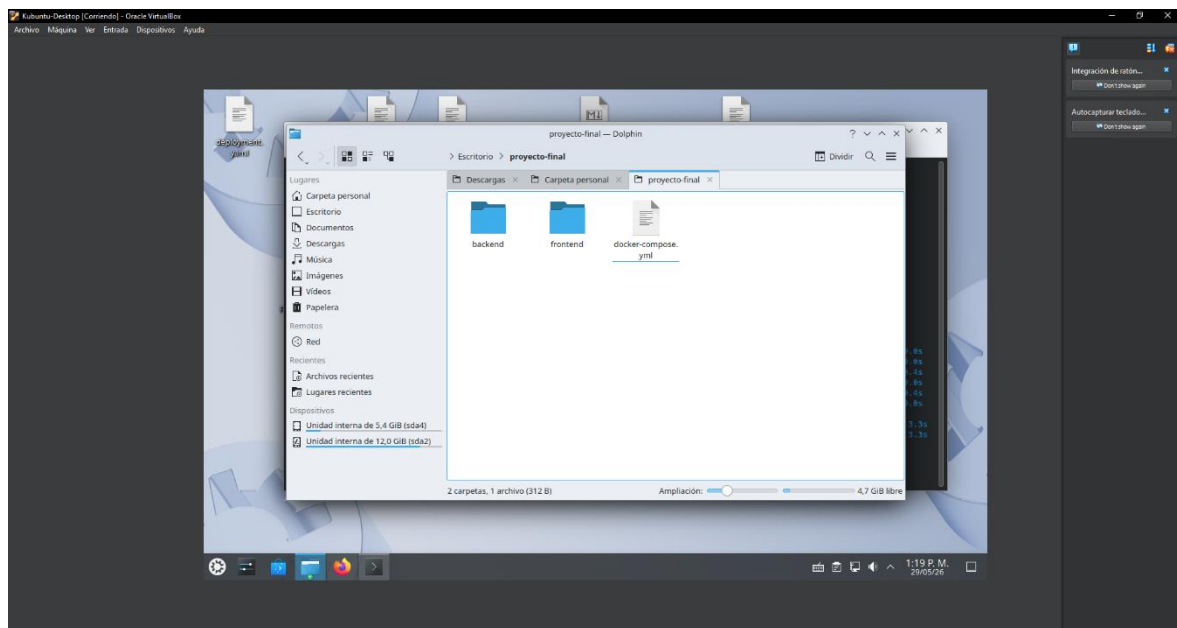


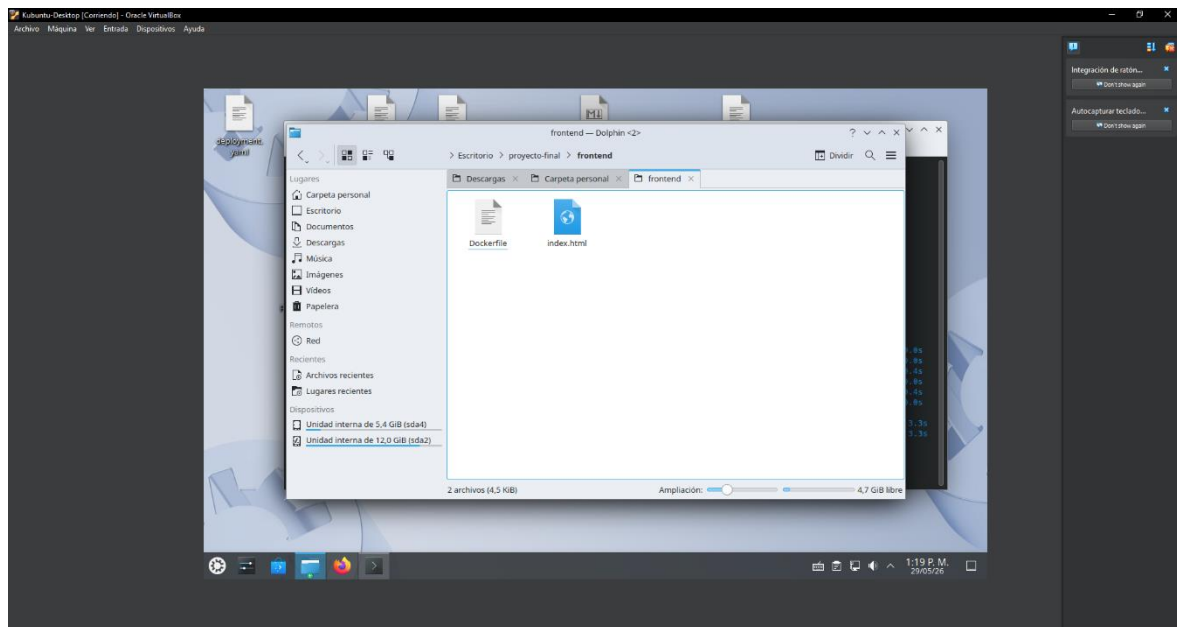
2.8 crear las carpetas back y front con “mkdir -p proyecto-final/frontend” y “mkdir -p proyecto-final/backend”, luego se crea Docker file tanto frontend como backend, para crear “touch Dockerfile.frontend/ Dockerfile.backend” y para modificar con “nano Dockerfile.frontend/ Dockerfile.backend”



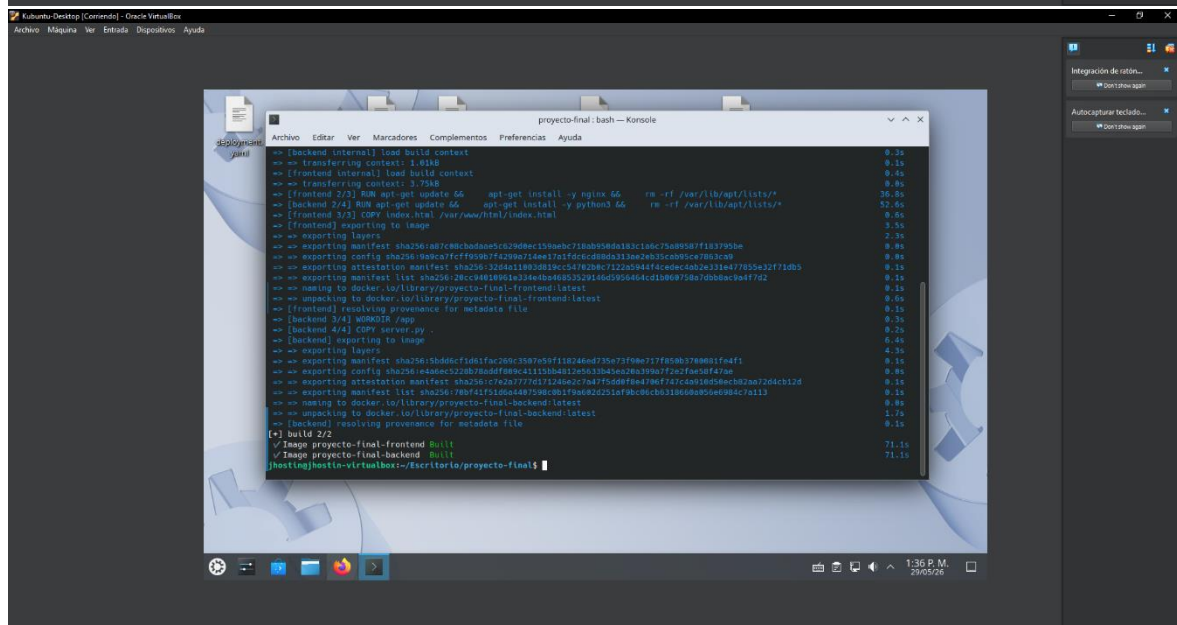
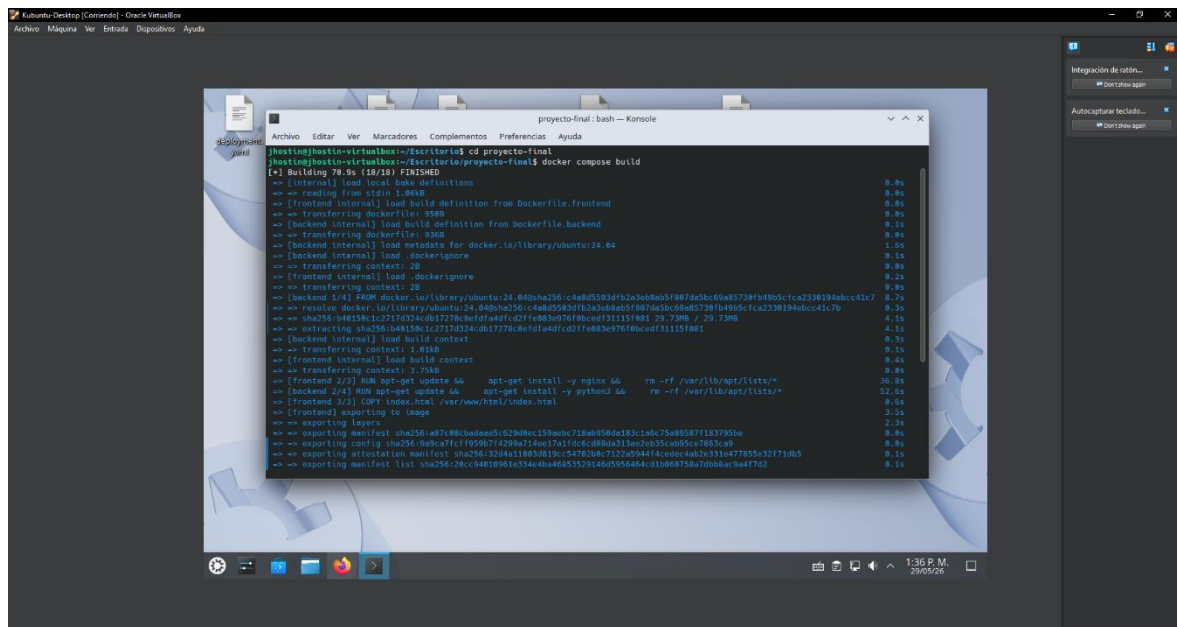


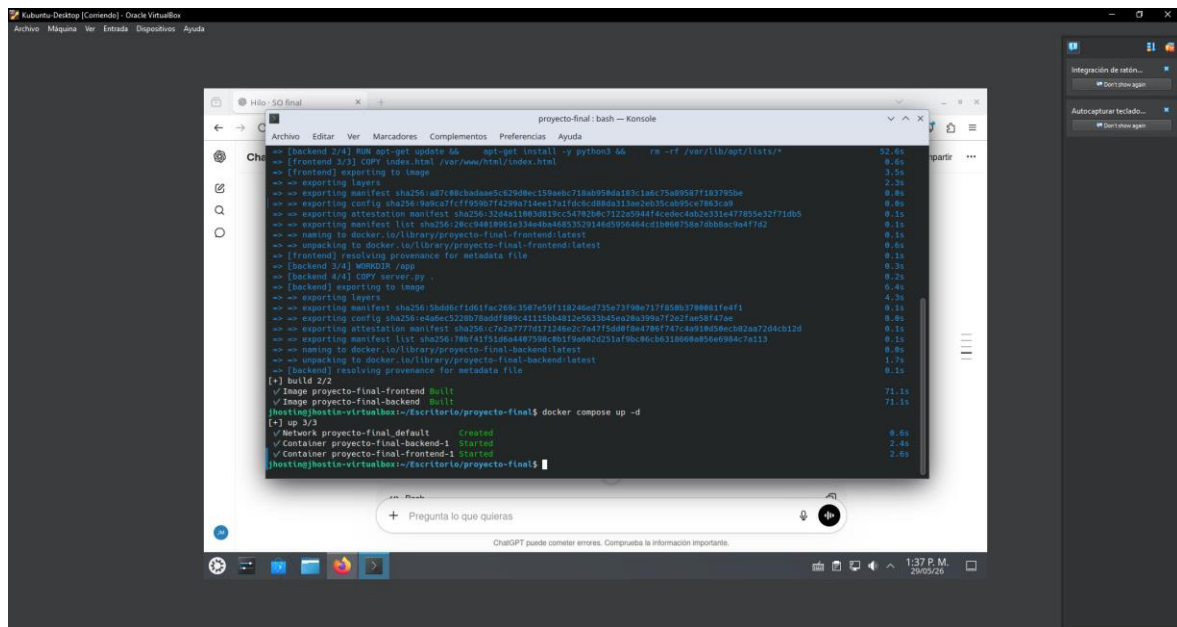
2.9 carpeta con todos los archivos



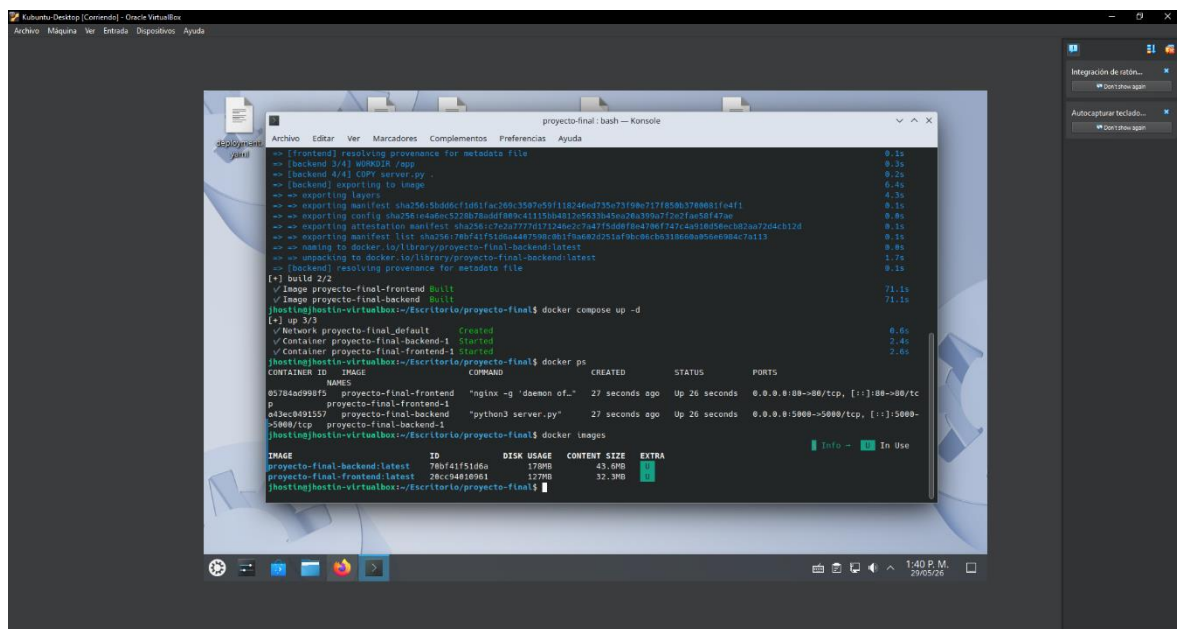


2.10 construir “docker compose build” y ejecutar “docker compose up -d” contenedores

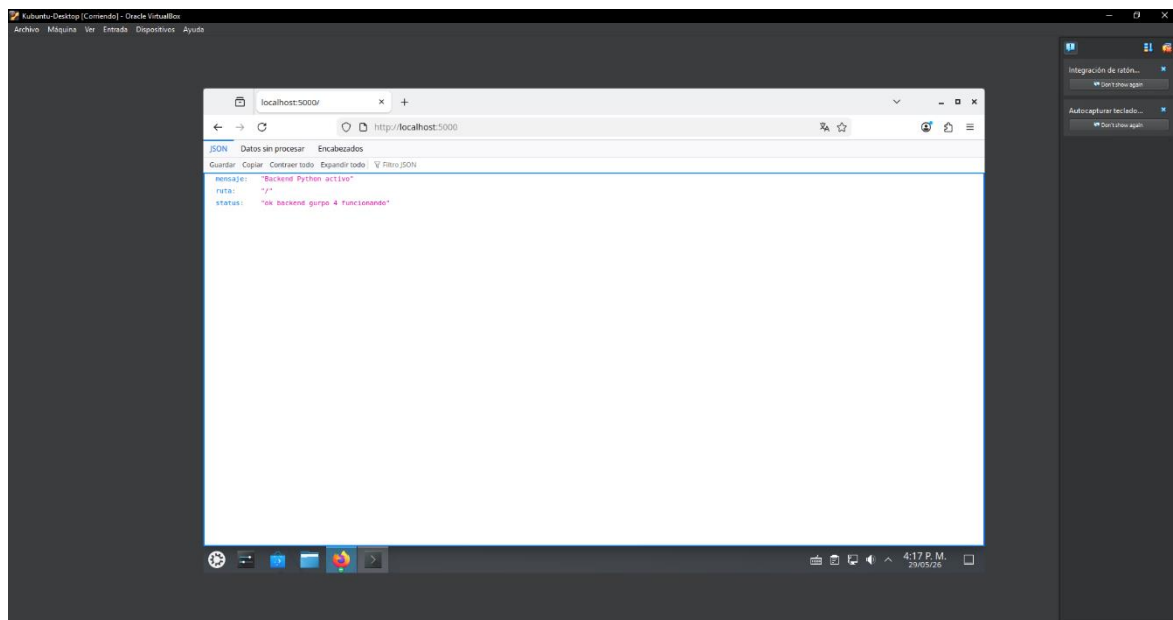
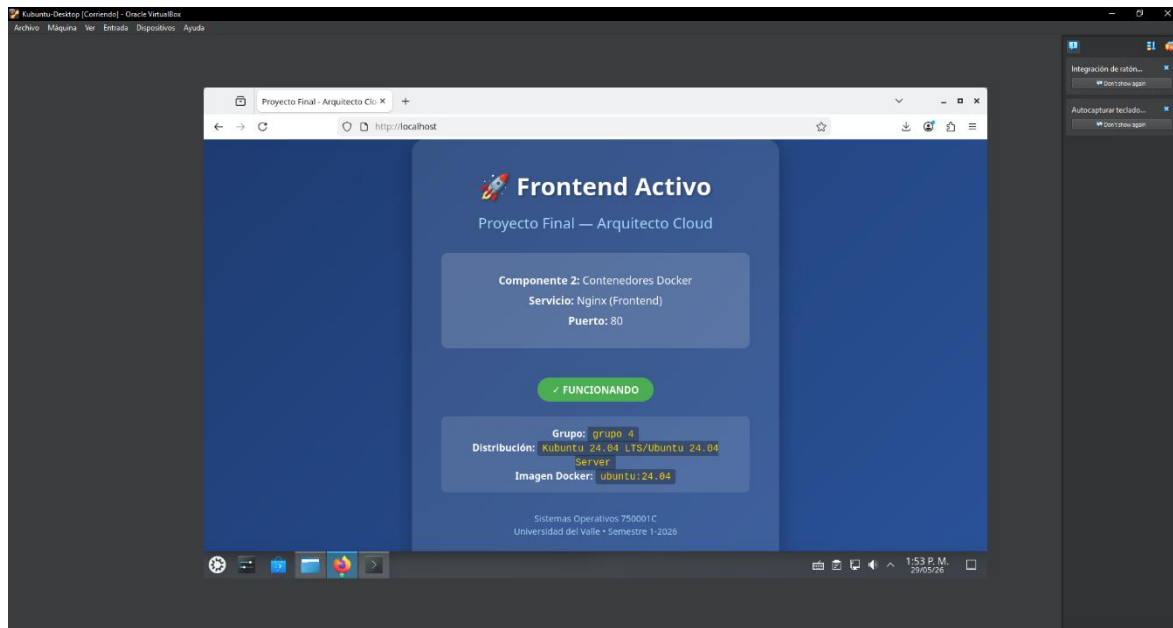




2.11 mostrar dockers con “docker ps” y “docker images”

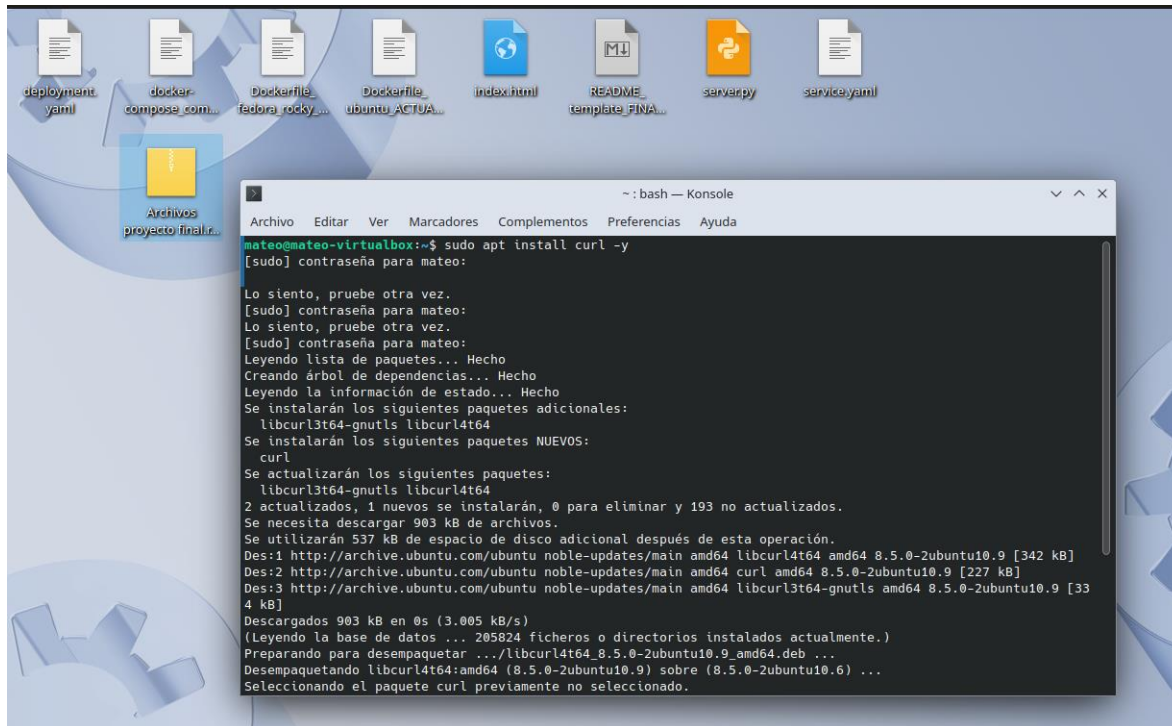


2.12 navegador accediendo al frontend y backend

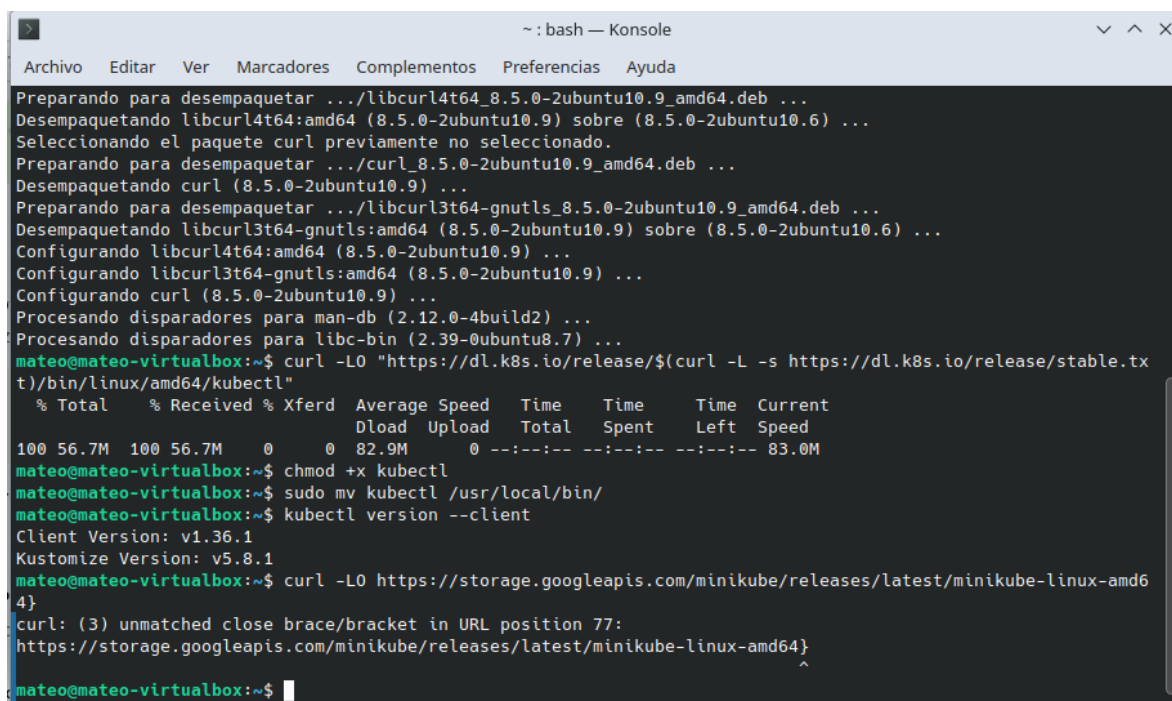


3. kubernets:

3.1 instalar kubectl para darle ordenes a los kubenets con “sudo apt install curl -y”, “curl -LO “https://dl.k8s.io/release/\$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl””, “chmod +x kubectl” y “sudo mv kubectl /usr/local/bin/”, luego de verifica la version con “kubectl version --client”



```
mateo@mateo-virtualbox:~$ sudo apt install curl -y
[sudo] contraseña para mateo:
Lo siento, pruebe otra vez.
[sudo] contraseña para mateo:
Lo siento, pruebe otra vez.
[sudo] contraseña para mateo:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
  libcurl3t64-gnutls libcurl4t64
Se instalarán los siguientes paquetes NUEVOS:
  curl
Se actualizarán los siguientes paquetes:
  libcurl3t64-gnutls libcurl4t64
2 actualizados, 1 nuevos se instalarán, 0 para eliminar y 193 no actualizados.
Se necesita descargar 903 kB de archivos.
Se utilizarán 537 kB de espacio de disco adicional después de esta operación.
Des:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.9 [342 kB]
Des:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.9 [227 kB]
Des:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.9 [33
4 kB]
Descargados 903 kB en 0s (3.005 kB/s)
(Leyendo la base de datos ... 205824 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../libcurl4t64_8.5.0-2ubuntu10.9_amd64.deb ...
Desempaquetando libcurl4t64:amd64 (8.5.0-2ubuntu10.9) sobre (8.5.0-2ubuntu10.6) ...
Seleccionando el paquete curl previamente no seleccionado.
```



```
mateo@mateo-virtualbox:~$ curl -LO https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.tx
t)/bin/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 56.7M 100 56.7M 0 0 82.9M 0 --:--:-- --:--:-- --:--:-- 83.0M
mateo@mateo-virtualbox:~$ chmod +x kubectl
mateo@mateo-virtualbox:~$ sudo mv kubectl /usr/local/bin/
mateo@mateo-virtualbox:~$ kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1
mateo@mateo-virtualbox:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd6
4}
curl: (3) unmatched close brace/bracket in URL position 77:
https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64}
^
mateo@mateo-virtualbox:~$
```

3.2 Instalar Minikube “curl -LO <https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64>”, “sudo install minikube-linux-amd64 /usr/local/bin/minikube” y verificar versión “minikube version”

```
~: bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda

drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Imágenes
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Música
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Plantillas
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Público
drwx----- 3 mateo mateo 4,0K may 29 18:24 snap
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Vídeos
mateo@mateo-virtualbox:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 128M  100 128M    0     0  48.0M      0  0:00:02  0:00:02 --:--:-- 48.0M
mateo@mateo-virtualbox:~$ ls -lh
total 129M
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:25 Descargas
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Documentos
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:26 Escritorio
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Imágenes
-rw-rw-r-- 1 mateo mateo 129M may 29 18:39 minikube-linux-amd64
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Música
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Plantillas
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Público
drwx----- 3 mateo mateo 4,0K may 29 18:24 snap
drwxr-xr-x 2 mateo mateo 4,0K may 29 18:22 Vídeos
mateo@mateo-virtualbox:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
mateo@mateo-virtualbox:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
mateo@mateo-virtualbox:~$
```

3.3 iniciar Kubernetes “minikube start” y verificar que funciona “kubectl get nodes”

```
~: bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda

mateo@mateo-virtualbox:~$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
mateo@mateo-virtualbox:~$ sudo usermod -aG docker $USER
[sudo] contraseña para mateo:
mateo@mateo-virtualbox:~$ newgrp docker
mateo@mateo-virtualbox:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
mateo@mateo-virtualbox:~$ minikube start
🔧 minikube v1.38.1 en Ubuntu 24.04 (vbox/amd64)
👉 Controlador docker seleccionado automáticamente. Otras opciones: none, ssh
❗ Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
🔥 Using Docker driver with root privileges
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📥 Pulling base image v0.0.50 ...
📦 Descargando Kubernetes v1.35.1 ...
> preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 54.02 M
> gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 23.89 M
🔥 Creating docker container (CPUs=2, Memory=3072MB) ...
👉 Preparando Kubernetes v1.35.1 en Docker 29.2.1...
🔧 Configurando CNI bridge CNI ...
🔧 Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Complementos habilitados: default-storageclass, storage-provisioner
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
mateo@mateo-virtualbox:~$ kubectl get nodes
NAME          STATUS    ROLES          AGE    VERSION
minikube      Ready     control-plane  3m38s  v1.35.1
mateo@mateo-virtualbox:~$
```

3.4 aplicar los yalm "kubectl apply -f deployment.yaml" y "kubectl apply -f service.yaml"

```
kubernetes: bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda

mateo@mateo-virtualbox:~$ cd Escritorio
mateo@mateo-virtualbox:~/Escritorio$ cd kubernetes
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/nginx created
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl apply -f service.yaml
service/nginx created
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-59f86b59ff-bcscm             1/1     Running   0           52s
nginx-59f86b59ff-dkgjj             1/1     Running   0           52s
mateo@mateo-virtualbox:~/Escritorio/kubernetes$
```

3.4 Verificar pods con "kubectl get pods" y verificar servicio con "kubectl get svc"

```
kubernetes: bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda

mateo@mateo-virtualbox:~$ cd Escritorio
mateo@mateo-virtualbox:~/Escritorio$ cd kubernetes
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/nginx created
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl apply -f service.yaml
service/nginx created
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-59f86b59ff-bcscm             1/1     Running   0           52s
nginx-59f86b59ff-dkgjj             1/1     Running   0           52s
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl get svc
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes ClusterIP  10.96.0.1     <none>        443/TCP          9m1s
nginx     NodePort  10.107.162.175 <none>        80:30080/TCP     60s
mateo@mateo-virtualbox:~/Escritorio/kubernetes$
```

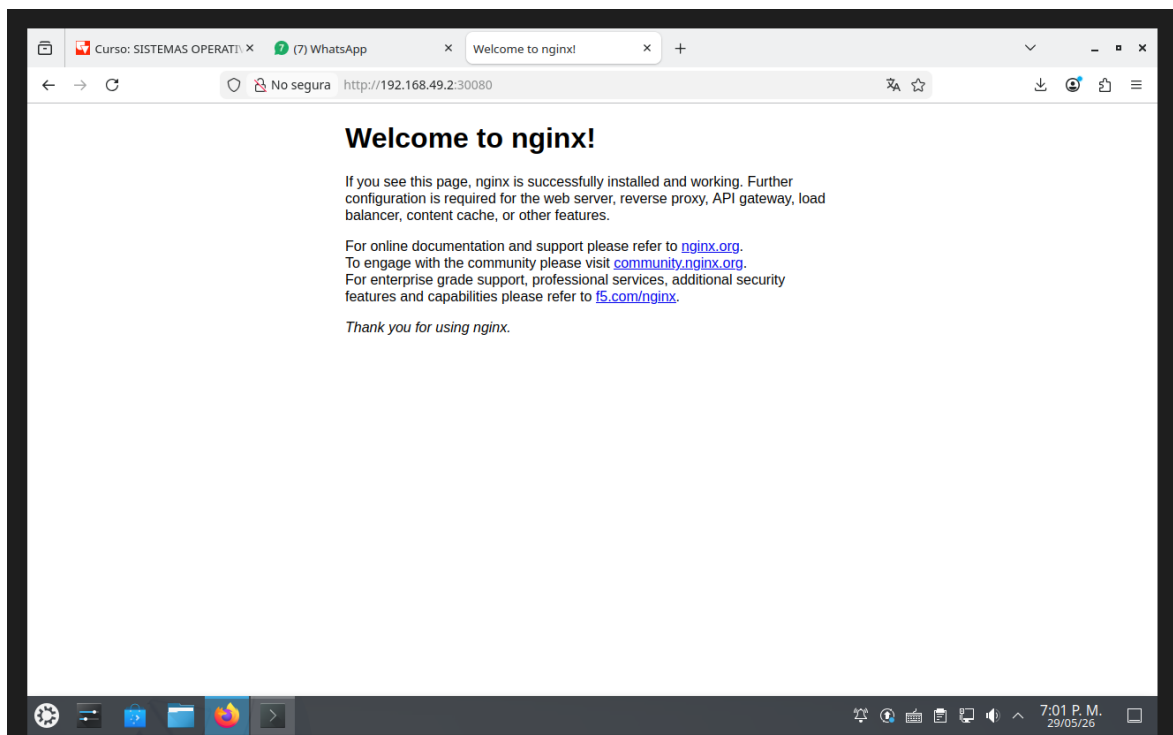
3.5 verificar URL “minikube service nginx --url”, antes de verificar la URL se tuvo que iniciar de nuevo el Docker

```

-: bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda

may 29 18:30:55 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:30:55.096438321-05:00" level=info msg="Co>
may 29 18:30:55 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:30:55.104509337-05:00" level=info msg="Da>
may 29 18:30:55 mateo-virtualbox systemd[1]: Started docker.service - Docker Application Container Engine.
may 29 18:30:55 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:30:55.104655982-05:00" level=info msg="AP>
may 29 18:45:08 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:08.798163266-05:00" level=info msg="im>
may 29 18:45:09 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:09.995179545-05:00" level=info msg="sb>
may 29 18:45:10 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:10.042115274-05:00" level=info msg="ig>
may 29 18:45:10 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:10.515164851-05:00" level=info msg="sb>
may 29 18:45:16 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:16.156547485-05:00" level=info msg="ig>
may 29 18:45:24 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:24.967814148-05:00" level=info msg="sb>
lines 1-27
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo systemctl start docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo systemctl enable docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo usermod -aG docker $USER
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ newgrp docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS
NAMES
660cd5de82de   gcr.io/k8s-minikube/kicbase:v0.0.50  "/usr/local/bin/entr..."  14 minutes ago  Up 14 minutes
127.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127
.0.0.1:32772->32443/tcp   minikube
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ minikube service nginx --url
http://192.168.49.2:30080
mateo@mateo-virtualbox:~/Escritorio/kubernetes$
```

3.6 Acceso desde navegador con <http://192.168.49.2:30080> que fue la URL que devolvió “minikube service nginx --url”



3.7 escalar a 3 réplicas con “kubectl scale deployment nginx --replicas=3” y verificar el escalado con “kubectl get pods”

```
- : bash — Konsole
Archivo  Editar  Ver  Marcadores  Complementos  Preferencias  Ayuda
may 29 18:45:10 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:10.515164851-05:00" level=info msg="sb>
may 29 18:45:16 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:16.156547485-05:00" level=info msg="ig>
may 29 18:45:24 mateo-virtualbox dockerd[4650]: time="2026-05-29T18:45:24.967814148-05:00" level=info msg="sb>
lines 1-27
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo systemctl start docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo systemctl enable docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ sudo usermod -aG docker $USER
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ newgrp docker
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS
NAMES
660cd5de82de   gcr.io/k8s-minikube/kicbase:v0.0.50  "/usr/local/bin/entr..."  14 minutes ago  Up 14 minutes
127.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127
.0.0.1:32772->32443/tcp  minikube
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ minikube service nginx --url
http://192.168.49.2:30080
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ ^C
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl scale deployment nginx --replicas=3
deployment.apps/nginx scaled
mateo@mateo-virtualbox:~/Escritorio/kubernetes$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-59f86b59ff-bcscm              1/1     Running   0           8m53s
nginx-59f86b59ff-dkgjj              1/1     Running   0           8m53s
nginx-59f86b59ff-ms92l              1/1     Running   0           25s
mateo@mateo-virtualbox:~/Escritorio/kubernetes$
```